PATHOGENS IN OUR ENVIRONMENT

BIOL 1308

JENIFER GIFFORD, PHD

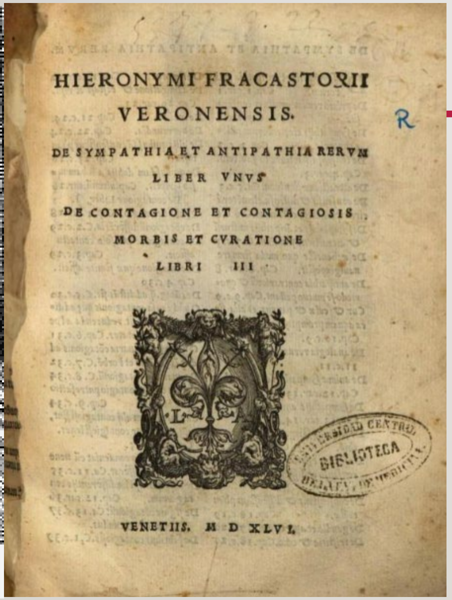
CILICIA NASCIMENTO, PHD

IN THIS LECTURE, WE WILL LEARN

1. What it means to be a pathogenic organism
2. The characteristics of viral, bacterial, protozoan, and fungal pathogens
3. How some pathogens are treated in the clinic

WHAT IS A PATHOGEN

MICROBIOLOGY TIMELINE



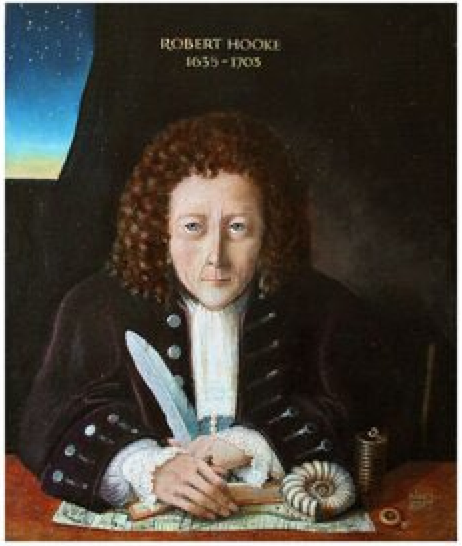
HIERONYMI FRACASTORII
VERONENSIS.
DE SYMPATHIA ET ANTIPATHIA RERVM
LIBER VNVS
DE CONTAGIONE ET CONTAGIOSIS
MORBIS ET CVRATIONE
LIBRI III
VENETIIS. M D XLVI.

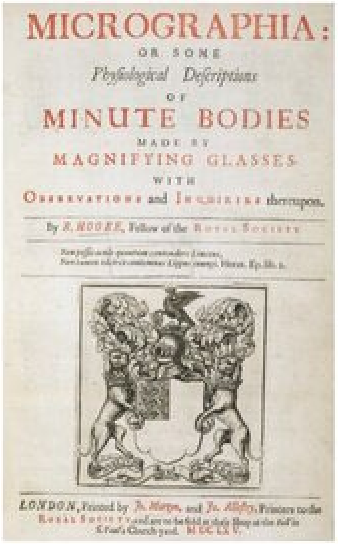
Girolamo Fracastoro

- Described a “contagion”

1546

1665



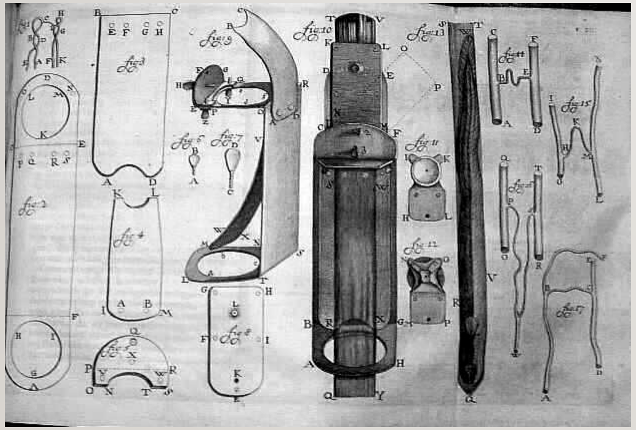


MICROGRAPHIA:
OR SOME
Physiological Descriptions
OF
MINUTE BODIES
MADE BY
MAGNIFYING GLASSES.
WITH
OBSERVATIONS and INQUIRIES thereupon.
By R. HOOKE, Fellow of the ROYAL SOCIETY.

Robert Hooke

- Investigated microscopic organisms

MICROBIOLOGY TIMELINE



Antoni van Leeuwenhoek

- First descriptions of bacteria

1676

Von den

Miasmen und Contagien

und von den

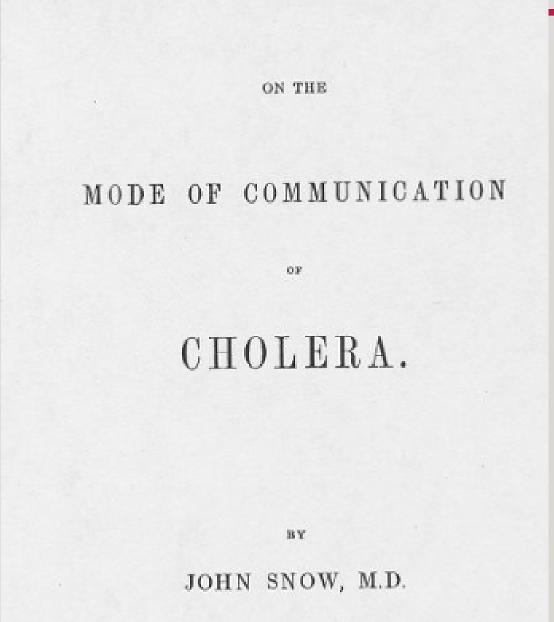
miasmatisch-contagiösen Krankheiten.

Friedrich Gustav Jakob Henle

- Germ theory of disease

1840

MICROBIOLOGY TIMELINE


ON THE

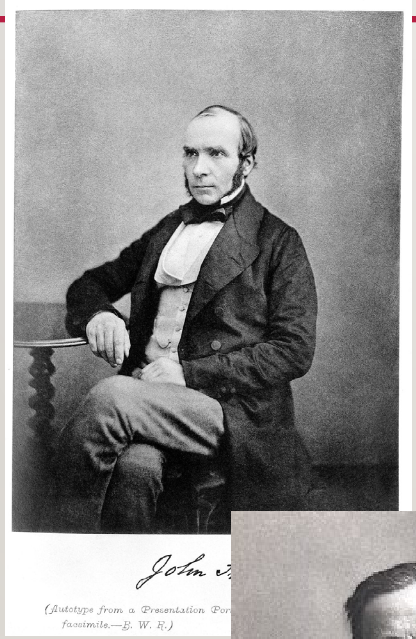
MODE OF COMMUNICATION

OF

CHOLERA.

BY

JOHN SNOW, M.D.



John Snow

- Published statistical evidence on cholera

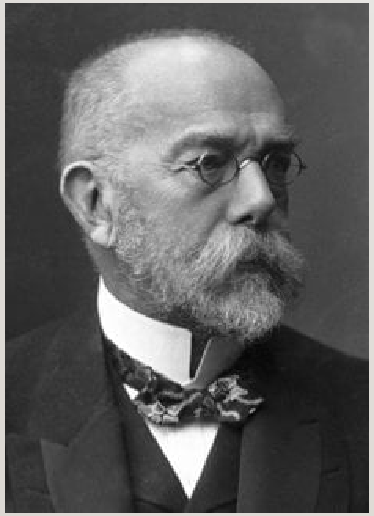
1854

Louis Pasteur

- Disproved spontaneous generation
- Pasteurization

1861

MICROBIOLOGY TIMELINE

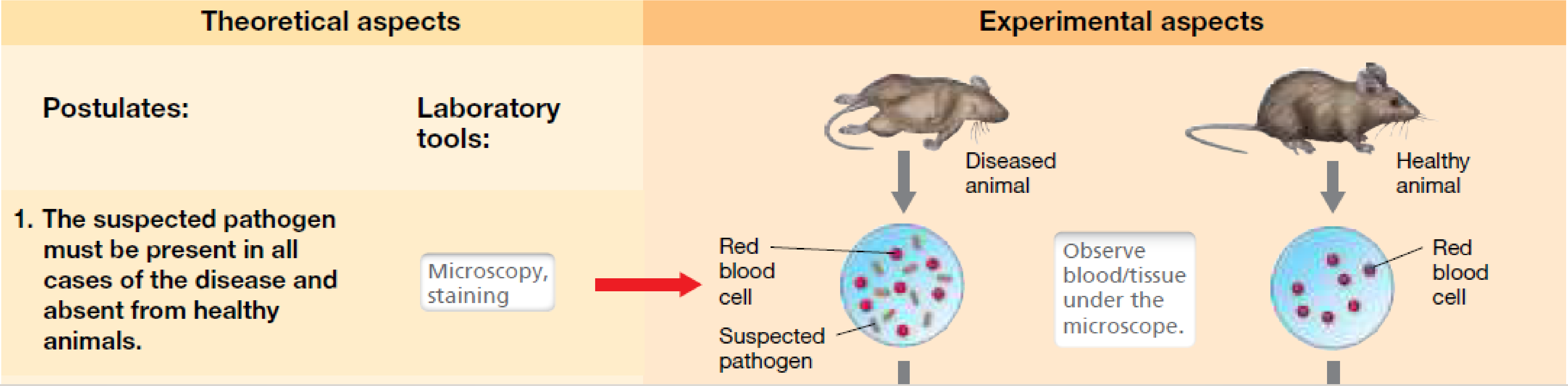


Robert Koch

- Established connection between Vibrio cholera and the disease
- Koch's Postulates

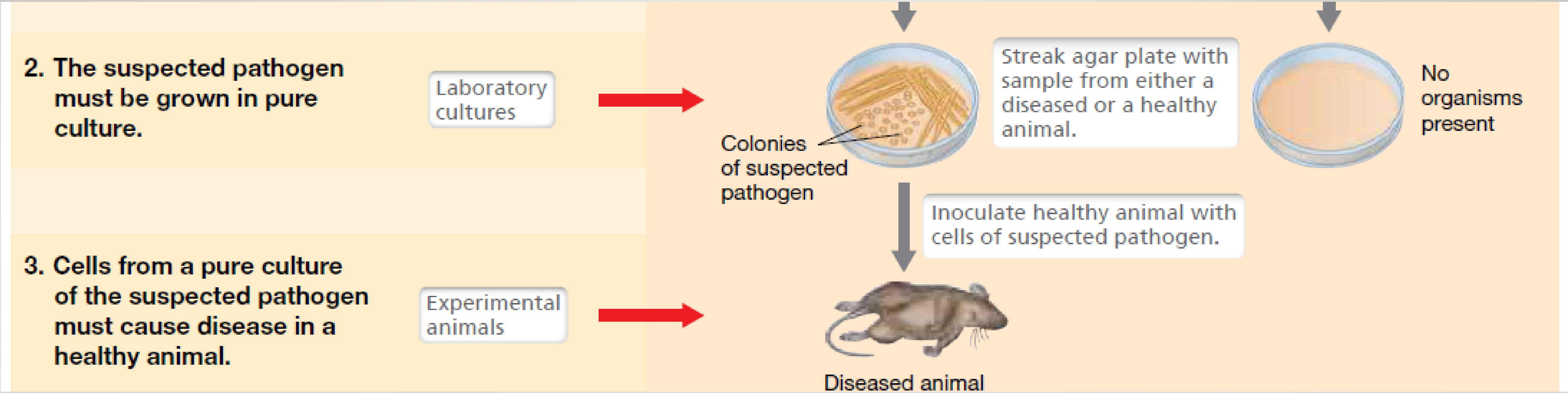
1884

KOCH'S POSTULATES



Pathogen:
A biological disease-causing organism or agent

KOCH'S POSTULATES



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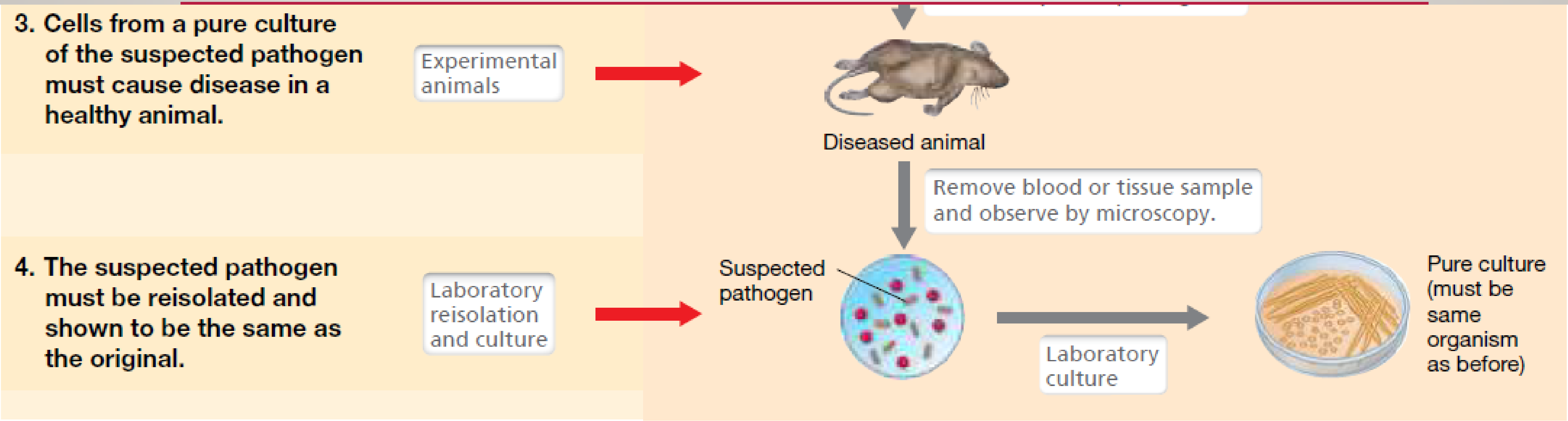
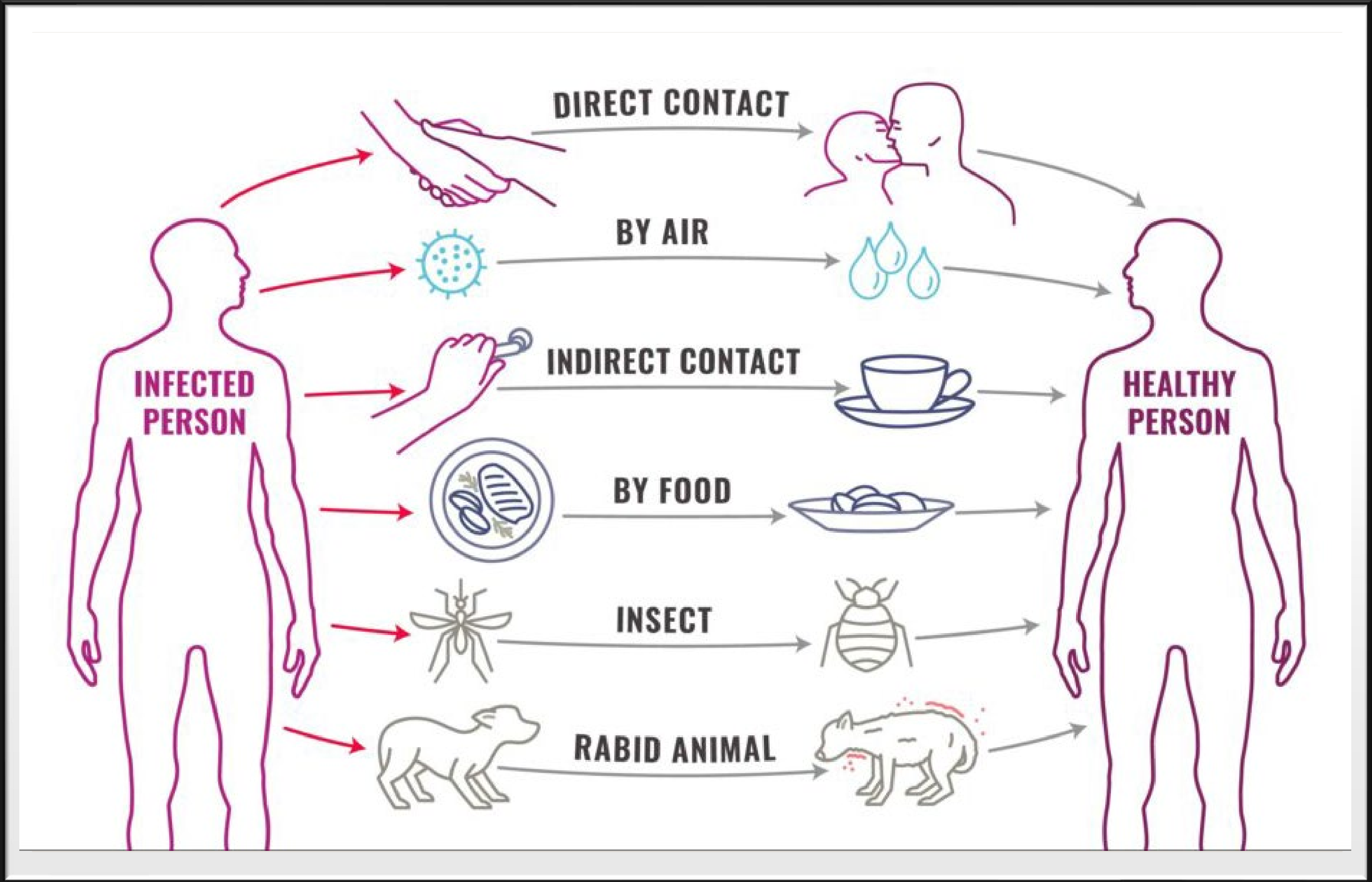


MODES OF TRANSMISSION



TYPES OF PATHOGENS

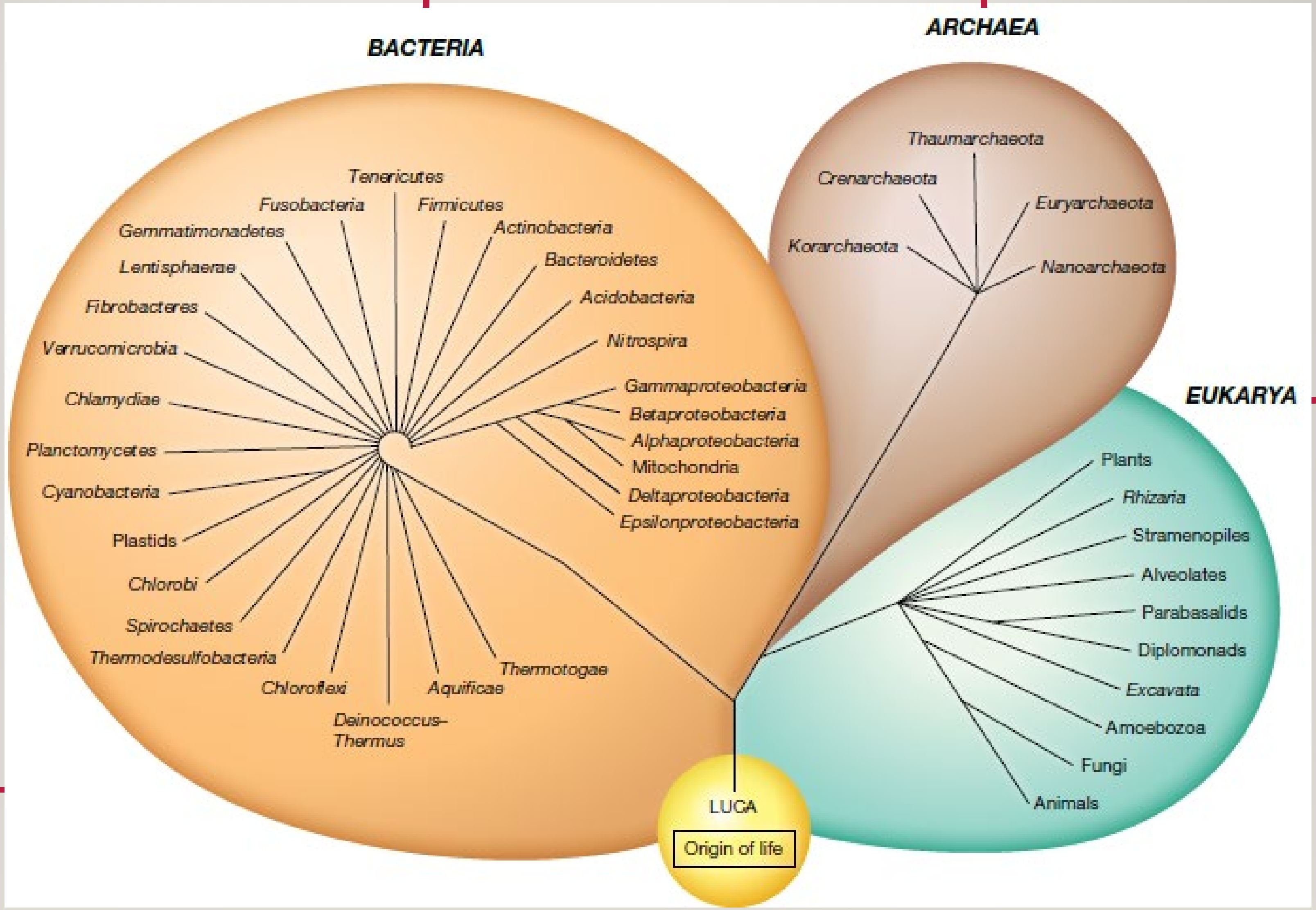
- Bacteria
  - Pneumonia
  - Food poisoning
  - Cholera
- Virus
  - Common cold
  - Flu
  - COVID-19
- Helminths

- Fungi
  - Athletes' food
  - Candidiasis
  - Ringworm
- Protists
  - Malaria
  - Giardiasis
  - Trichomoniasis

PHYLOGENY

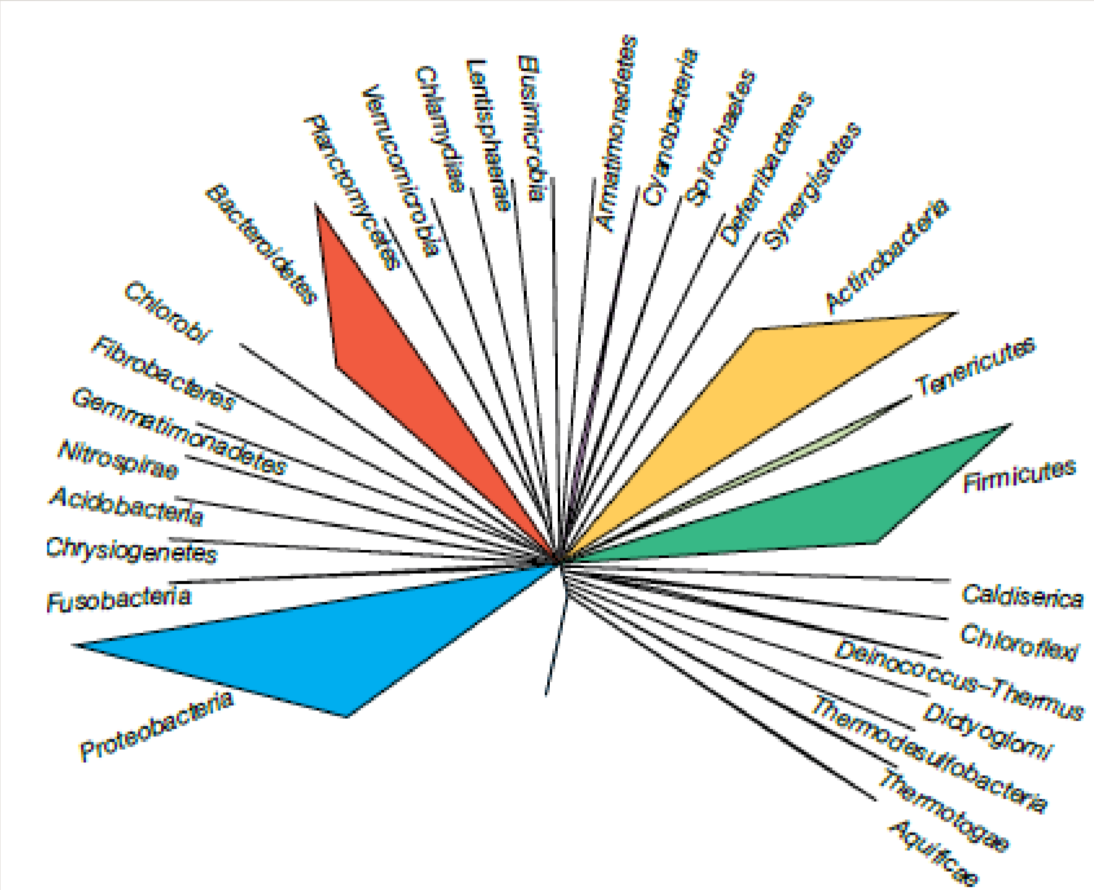
Prokaryotic

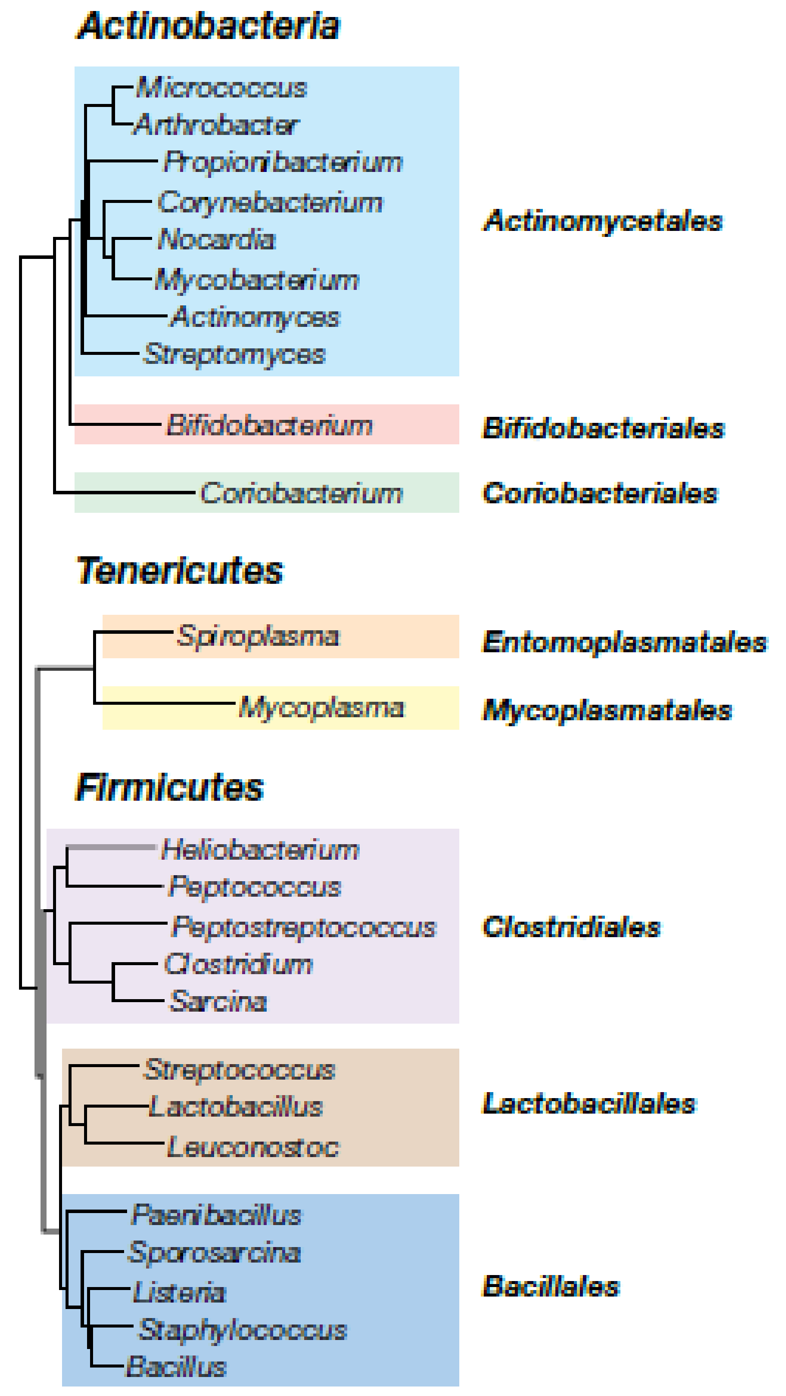
Eukaryotic



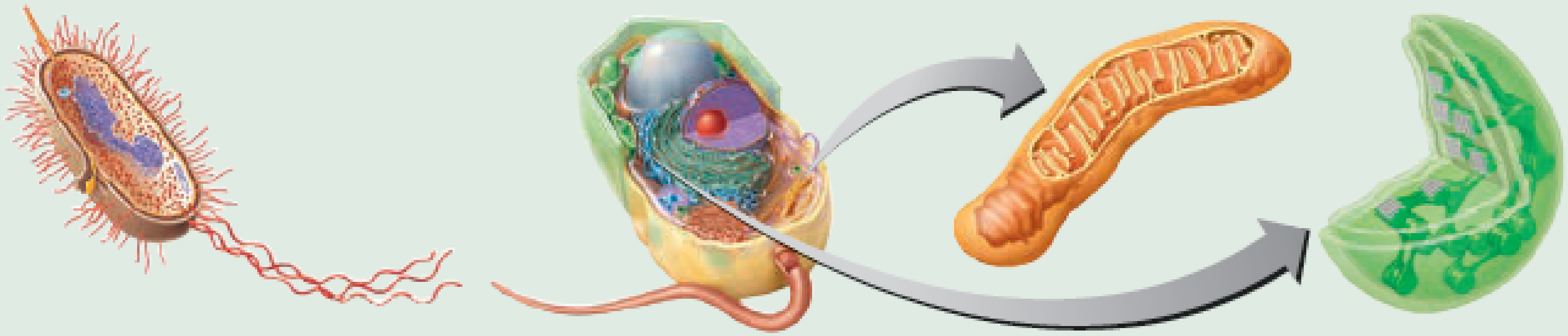
IMPORTANT FAMILIES OF BACTERIA

- **Firmicutes-** botulism
- **Actinobacteria**; Tuberculosis (TB)
- **Spirochaetes**: syphilis
- **Chlamydiae**: Chlamydia
- **Proteobacteria-** gonorrhea, e.coli, cholera, salmonella

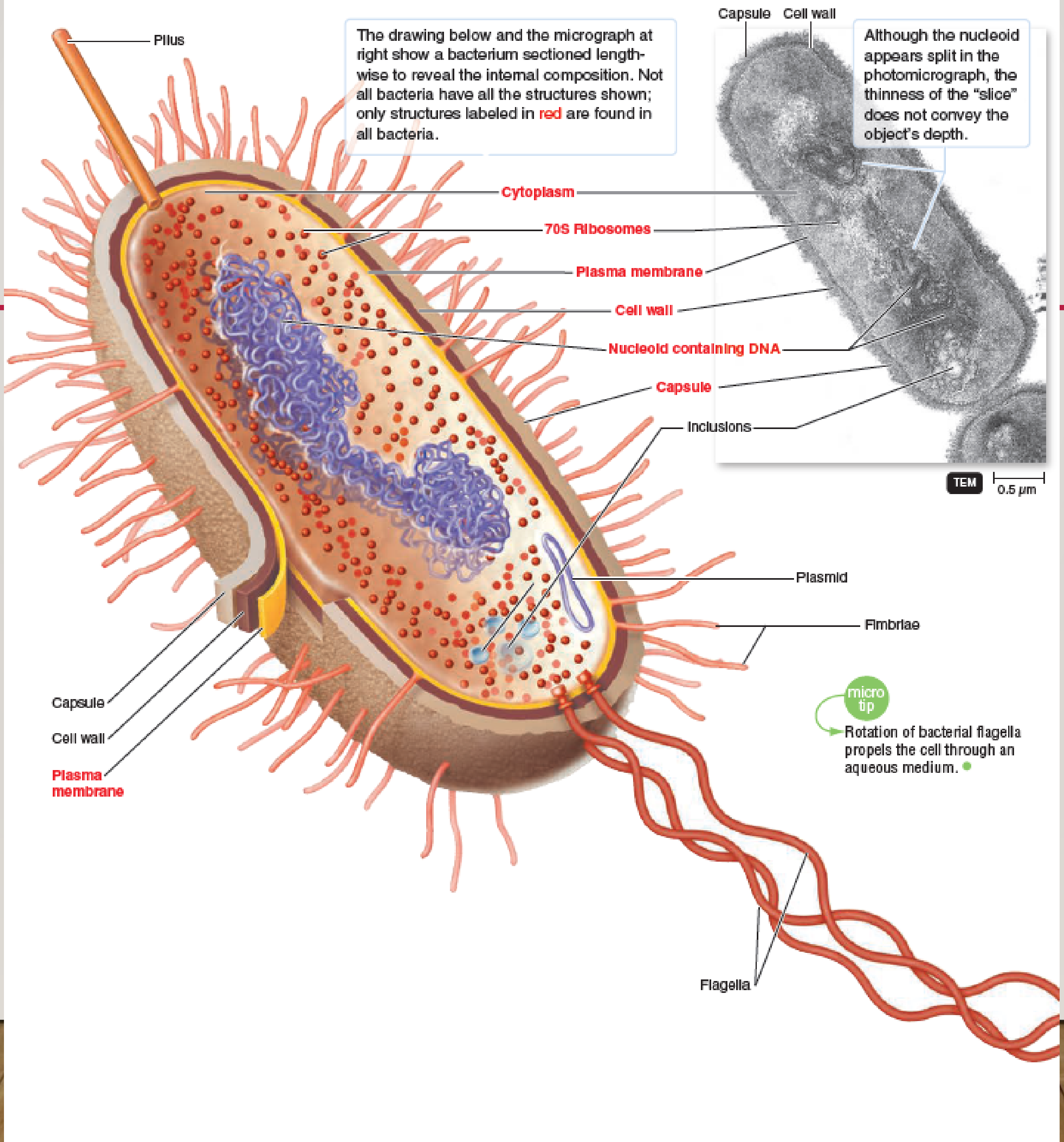




	Prokaryotic Cell	Eukaryotic Cell	Eukaryotic Organelles (Mitochondria and Chloroplasts)
DNA	One circular; some two circular; some linear	Linear	Circular
Histones	In archaea	Yes	No
First Amino Acid in Protein Synthesis	Formylmethionine (bacteria) Methionine (archaea)	Methionine	Formylmethionine
Ribosomes	70S	80S	70S
Growth	Binary fission	Mitosis	Binary fission



PROKARYOTIC MORPHOLOGY

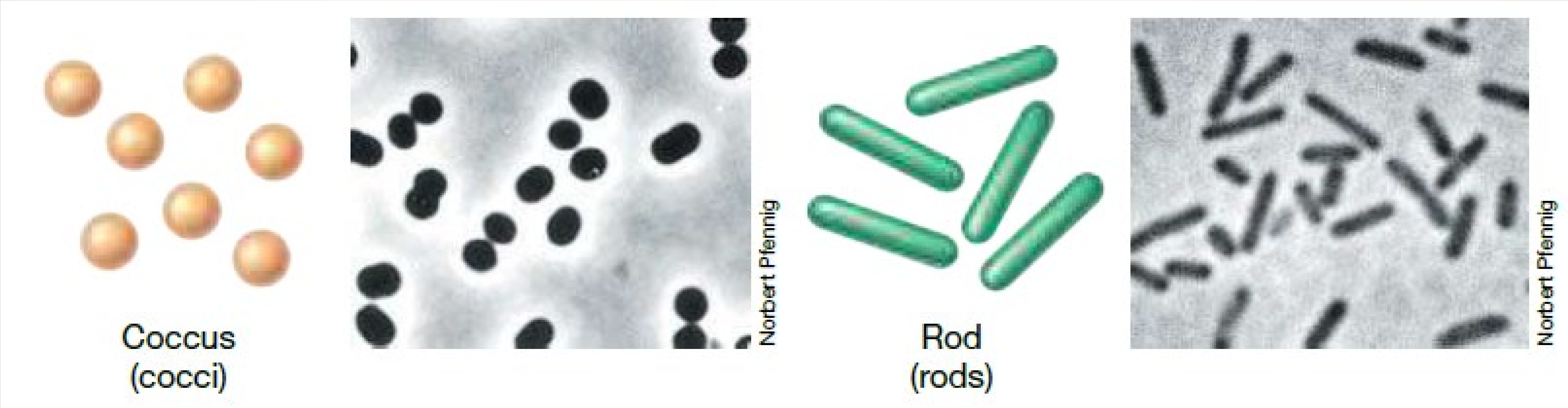


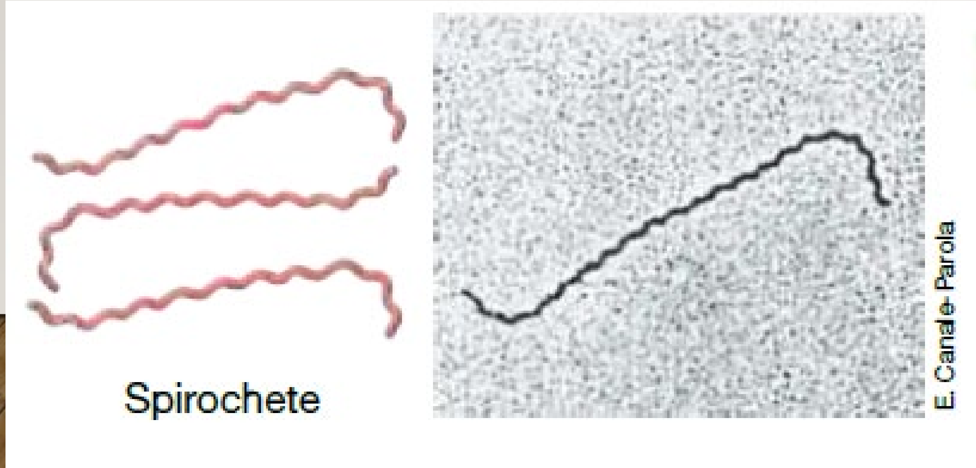
EXAMPLES OF PATHOGENIC BACTERIA

Lineage	Species	Tissues affected	Disease
Firmicutes	**Clostridium tetani** **Staphylococcus aureus** **Streptococcus pneumonia** **Streptococcus pyogenes**	Wounds, nervous system Skin, urogenital canal Respiratory tract Respiratory tract	Tetanus Acne, boils, impetigo, toxic shock syndrome Pneumonia Strep throat, scarlet fever
Spirochaetes	**Borrelia burgdorferi** **Treponema pallidum**	Skin and nerves Urogenital canal	Lyme disease Syphilis
Actinobacteria	**Mycobacterium tuberculosis** **Mycobacterium leprae** **Propionibacterium acnes**	Respiratory tract Skin and nerves Skin	Tuberculosis Leprosy Acne
Chlamydiales	**Chlamydia trachomatis**	Urogenital canal	Genital tract infection
ε-proteobacteria	**Helicobacter pylori**	Stomach	Ulcer
β-proteobacteria	**Neisseria gonorrhoeae**	Urogenital canal	Gonorrhea
γ-proteobacteria	**Haemophilus influenza** **Pseudomonas aeruginosa** **Salmonella enterica** **Yersinia pestis**	Ear canal, nervous system Urogenital canal, eyes, ear canal, lungs Gastrointestinal tract Lymph and blood	Ear infections, meningitis Infections of eye, ear, urinary tract, lungs Food poisoning Plague

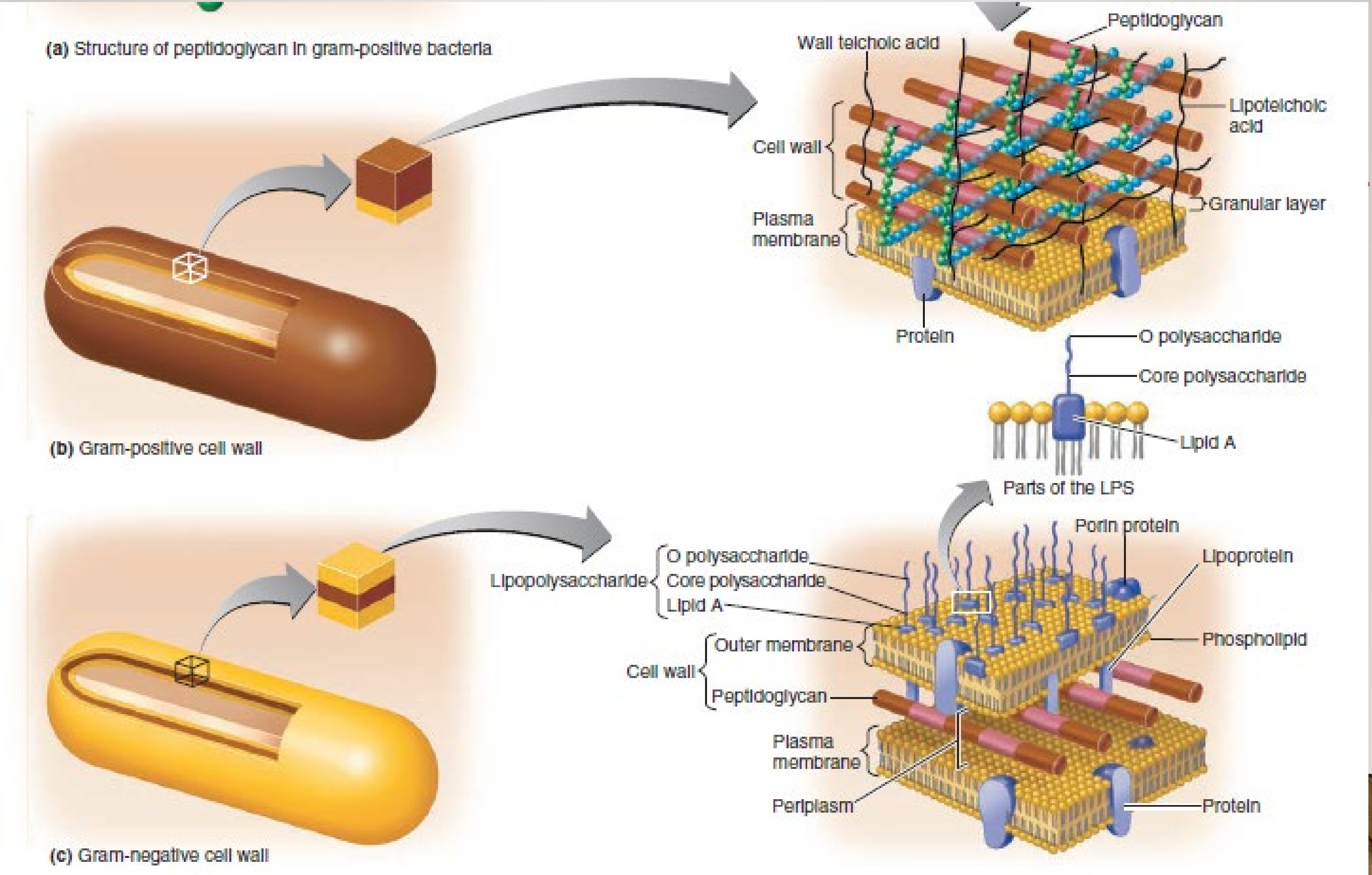
(do not memorize)

BACTERIAL STRUCTURE

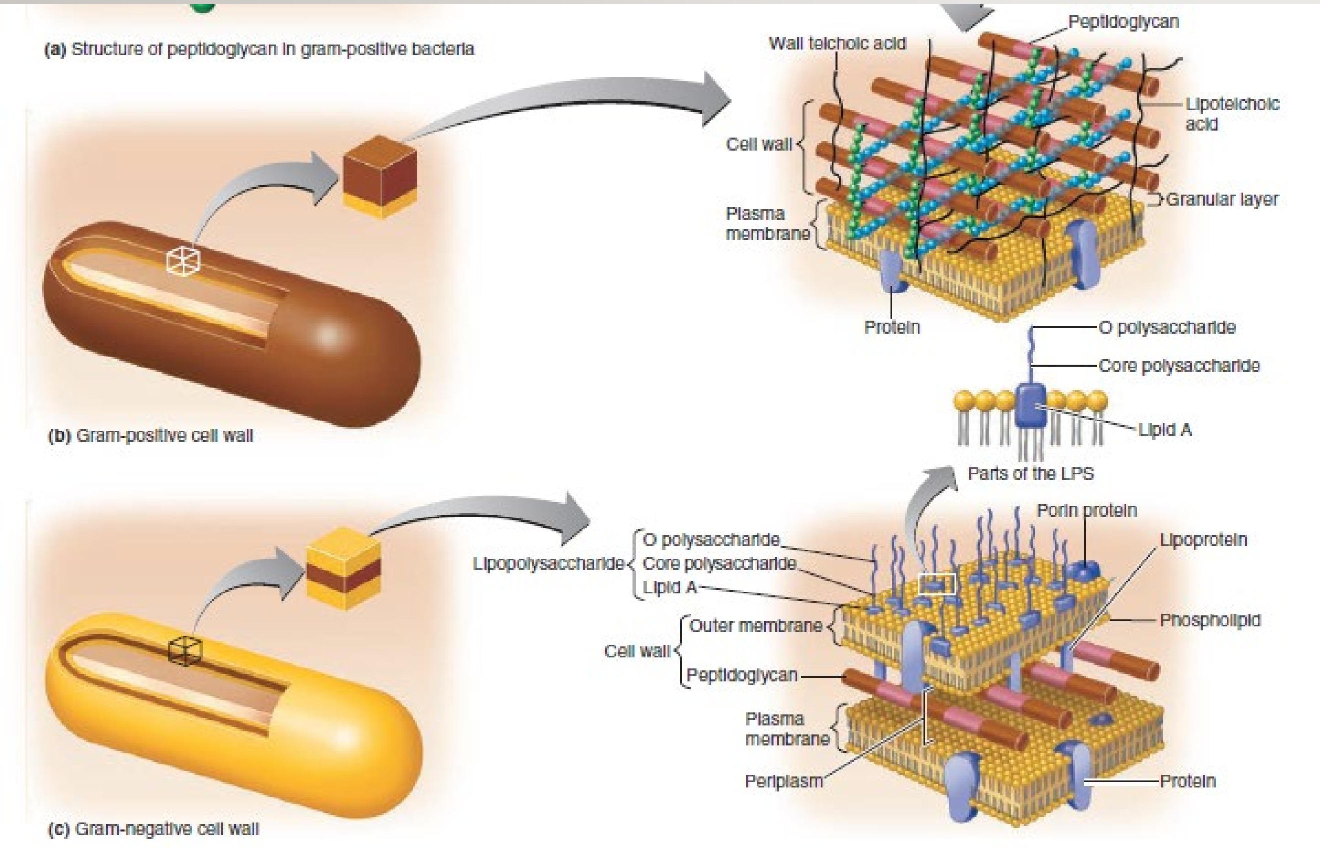




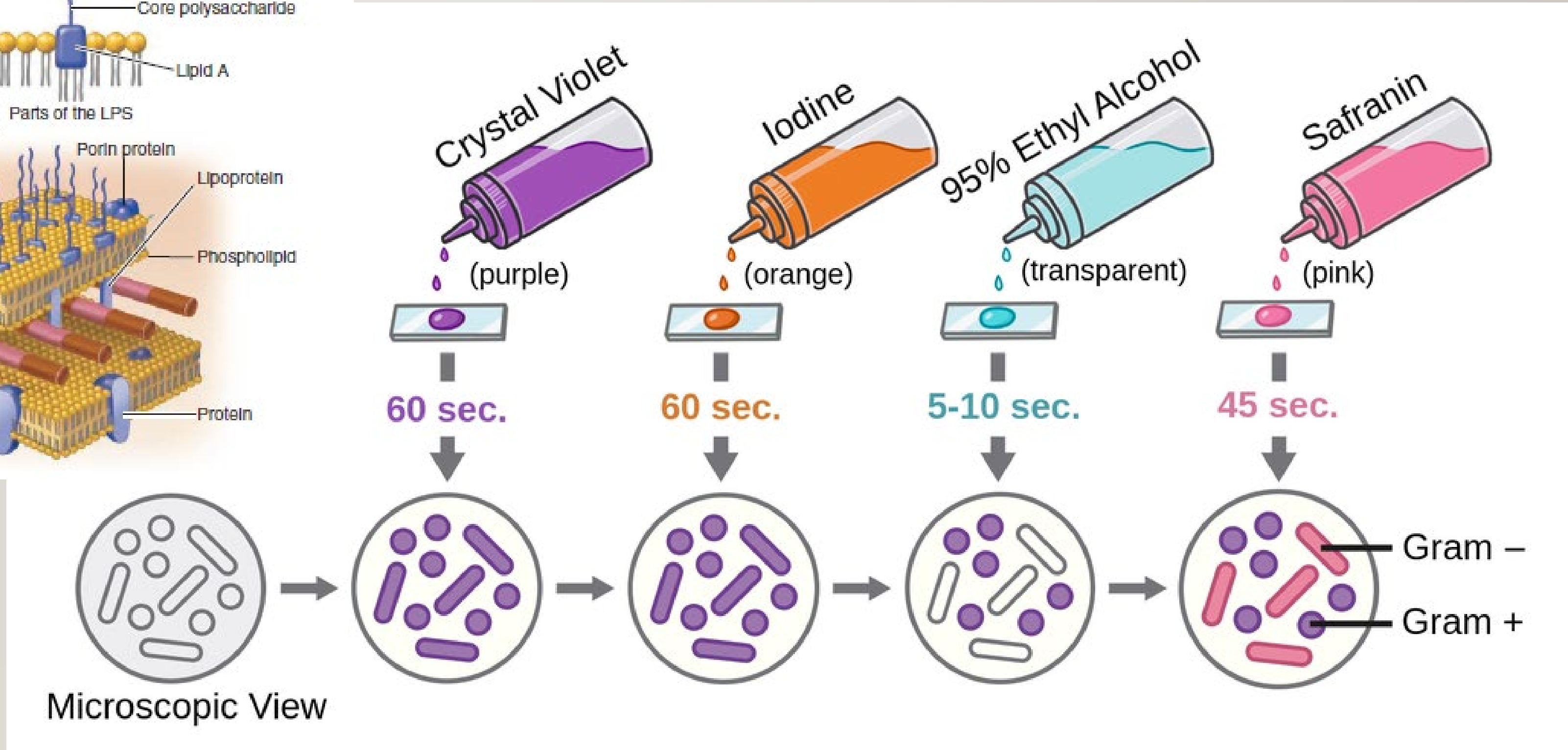
CELL WALL IS IMPORTANT



GRAM STAIN



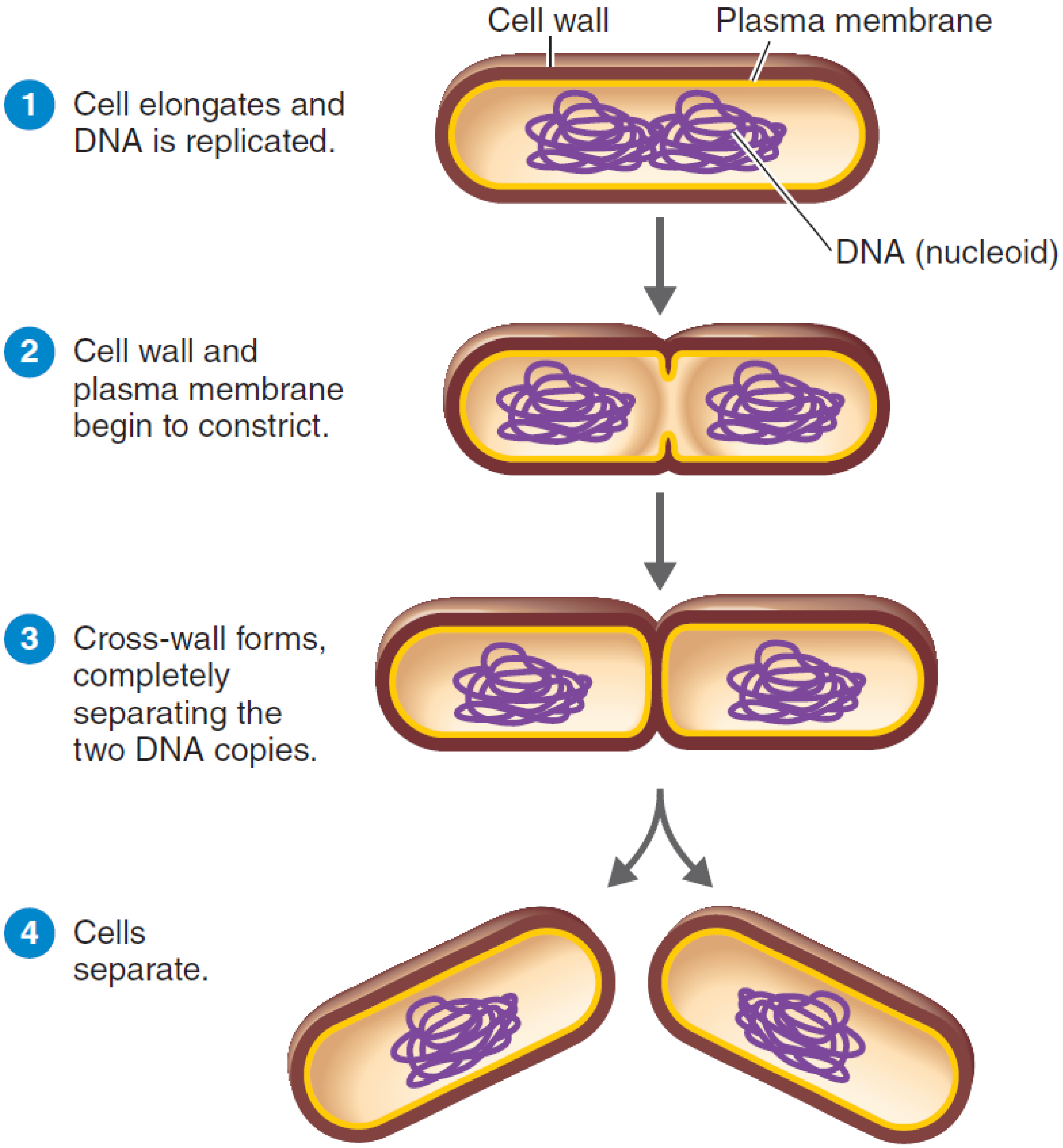
(a) Structure of peptidoglycan in gram-positive bacteria
Peptidoglycan
Wall teichoic acid
Lipoteichoic acid
Cell wall
Granular layer
Plasma membrane
Protein
(b) Gram-positive cell wall
O polysaccharide
Core polysaccharide
Lipid A
Parts of the LPS
Porin protein
Lipoprotein
Lipopolysaccharide
O polysaccharide
Core polysaccharide
Lipid A
Phospholipid
Outer membrane
Cell wall
Peptidoglycan
Plasma membrane
Periplasm
Protein
(c) Gram-negative cell wall



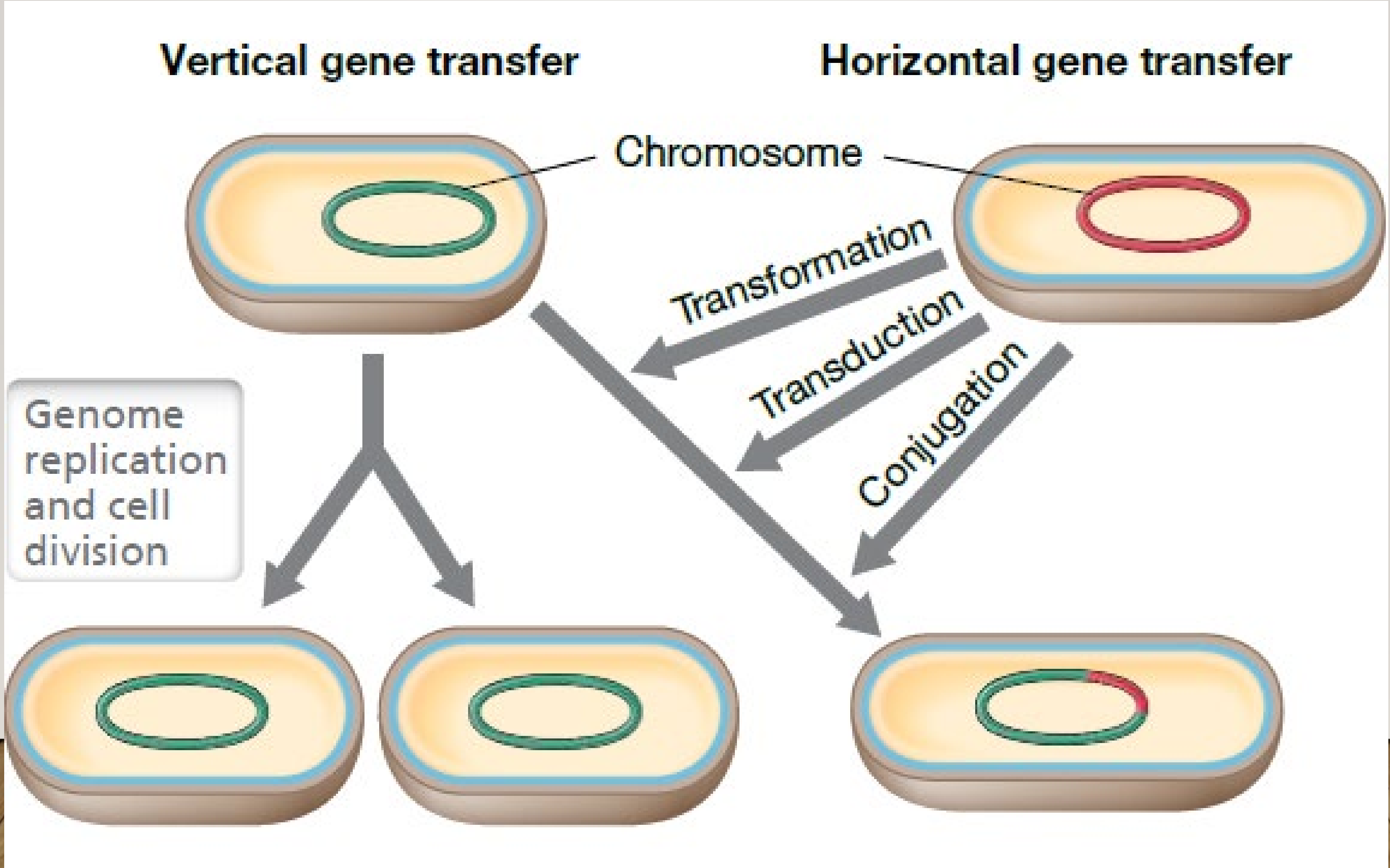
Crystal Violet
(purple)
60 sec.
Iodine
(orange)
60 sec.
95% Ethyl Alcohol
(transparent)
5-10 sec.
Safranin
(pink)
45 sec.
Gram –
Gram +
Microscopic View

BACTERIA AND ARCHAEA REPRODUCTION

- Asexual binary fission
- Generation time is as short as 12 min
- Mutations are generated rapidly and passed on to offspring more quickly than eukaryotes

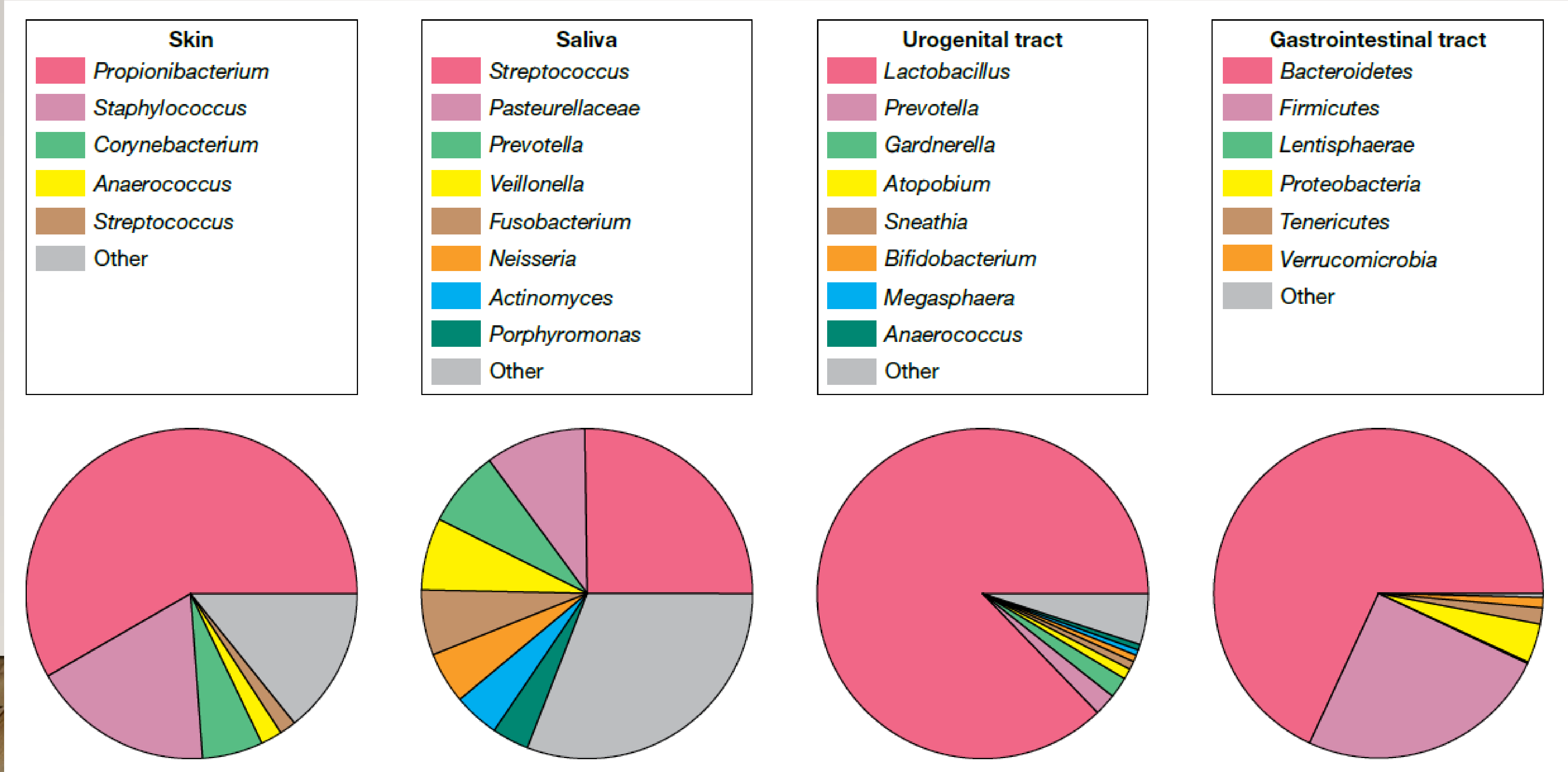


GENE TRANSFER



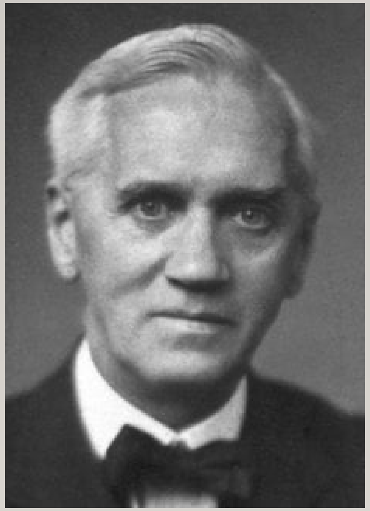
HUMAN MICROBIOME

Pathogen:
A biological disease-causing organism or agent

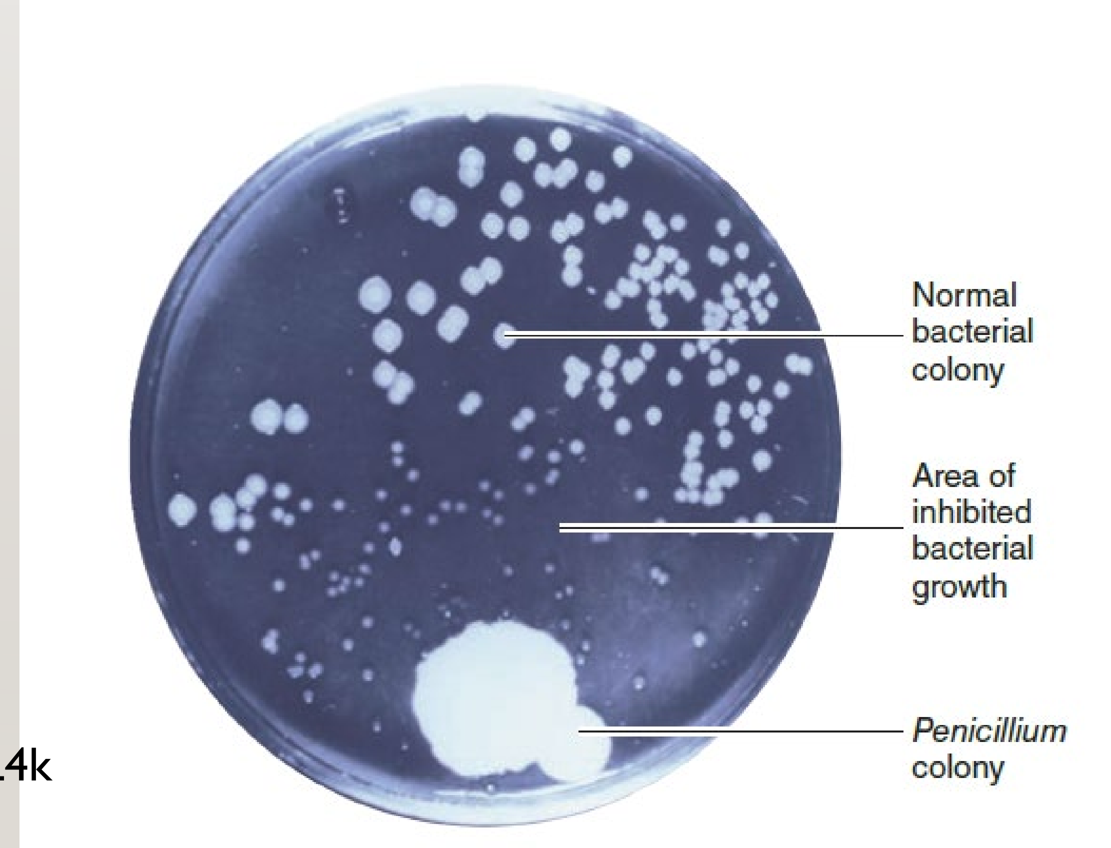


ANTIBIOTICS

- Alexander Fleming (1928)
  - Accidental discovery of mold that inhibited bacterial growth
  - *Penicillium chrysogenum*
  - penicillin

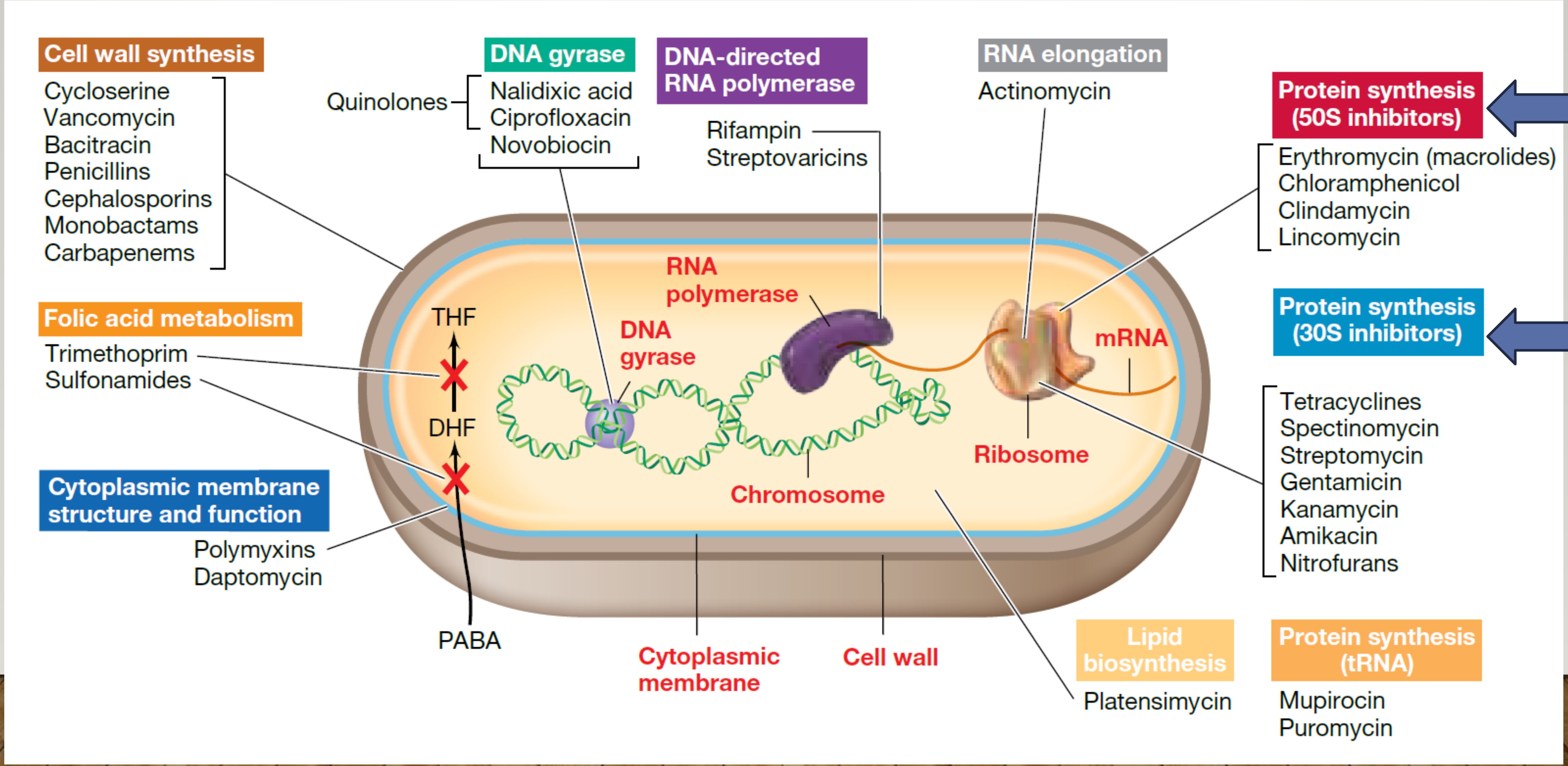


https://youtu.be/YOrRQtA8BsY?si=y3_VhUVLnC3QQL4k



ANTIMICROBIAL AGENTS

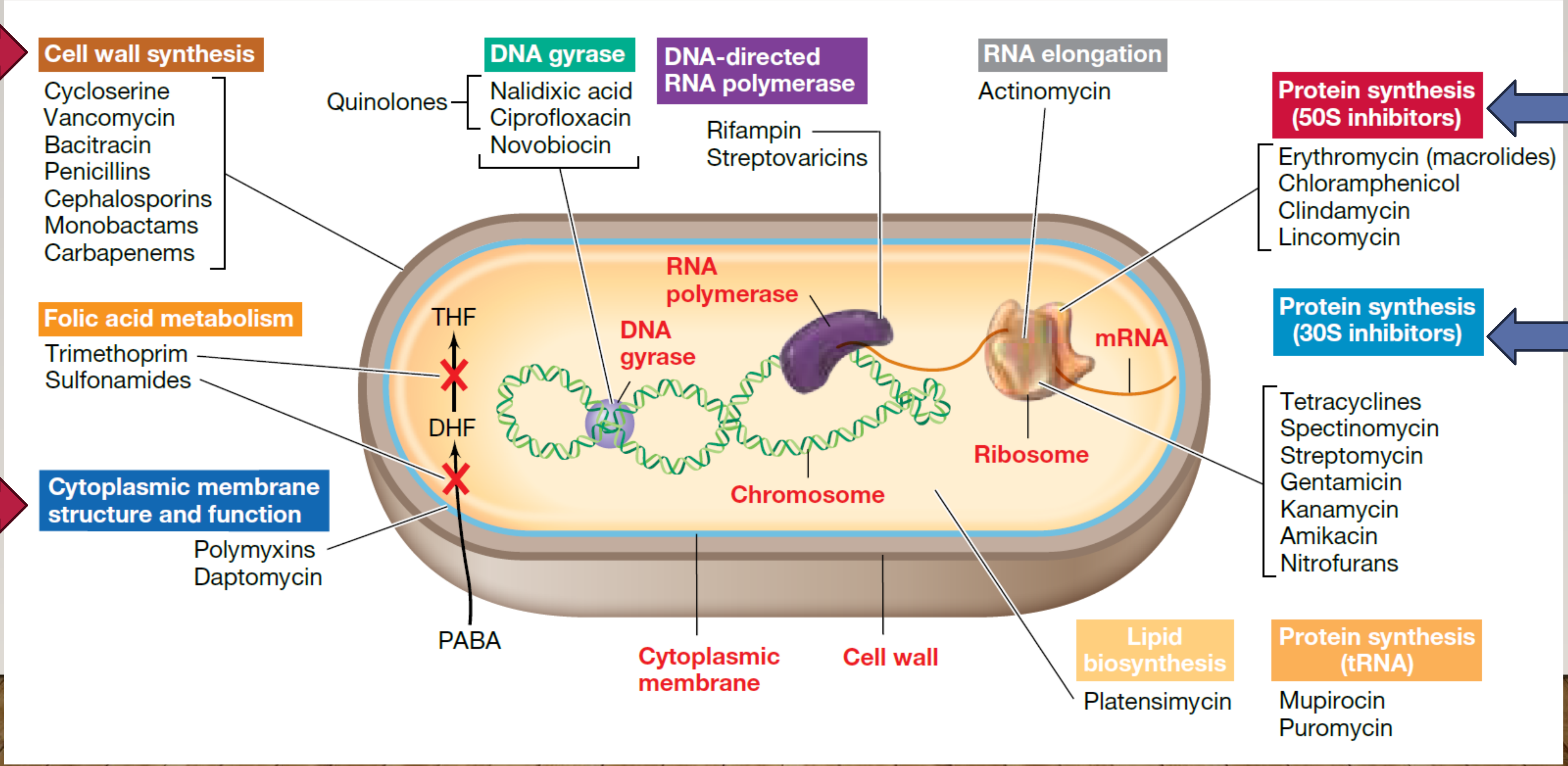
Bacteriostatic



ANTIMICROBIAL AGENTS

Bacteriostatic

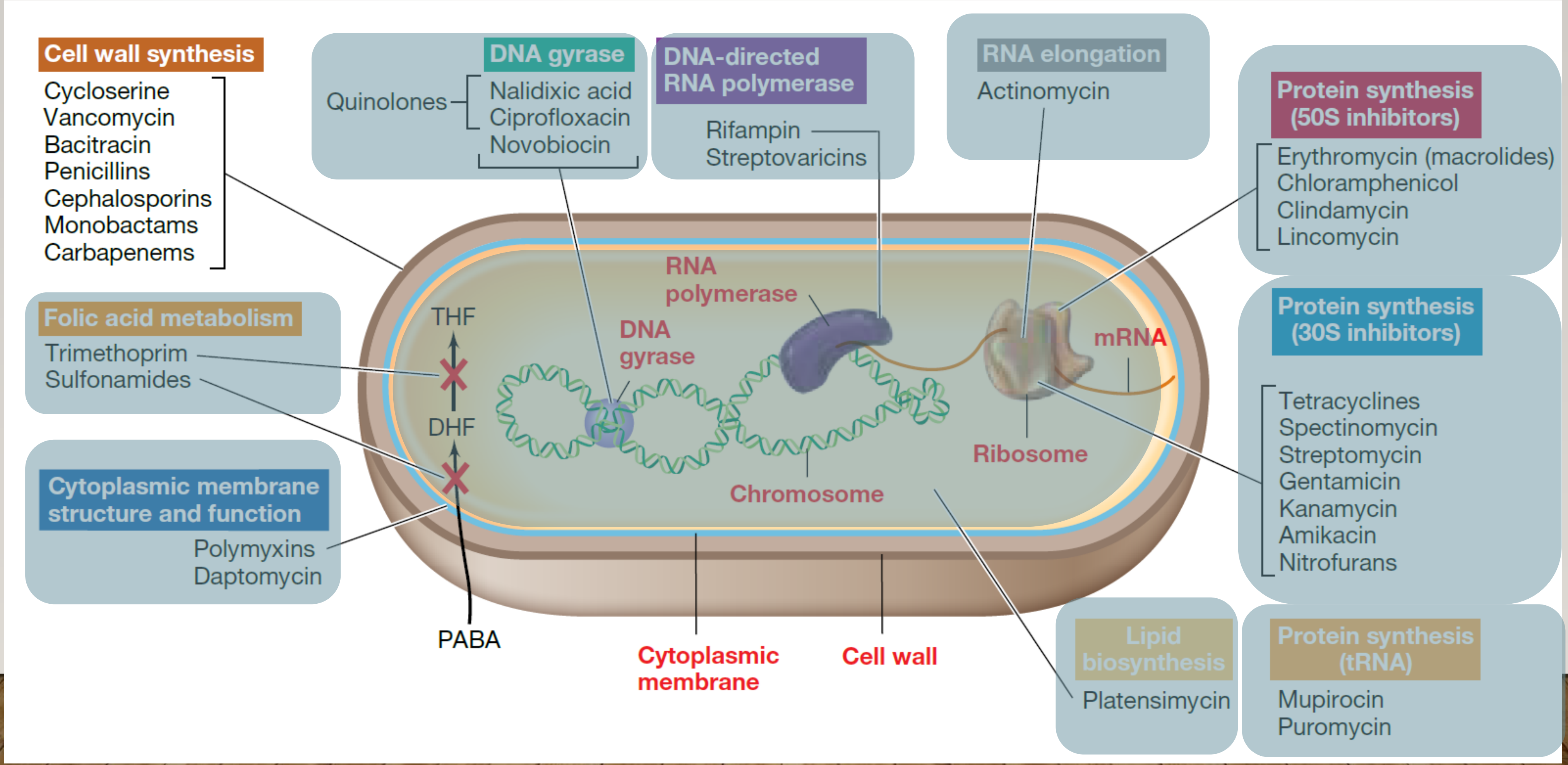
Bactericidal



ANTIMICROBIAL AGENTS

Internal protein synthesis

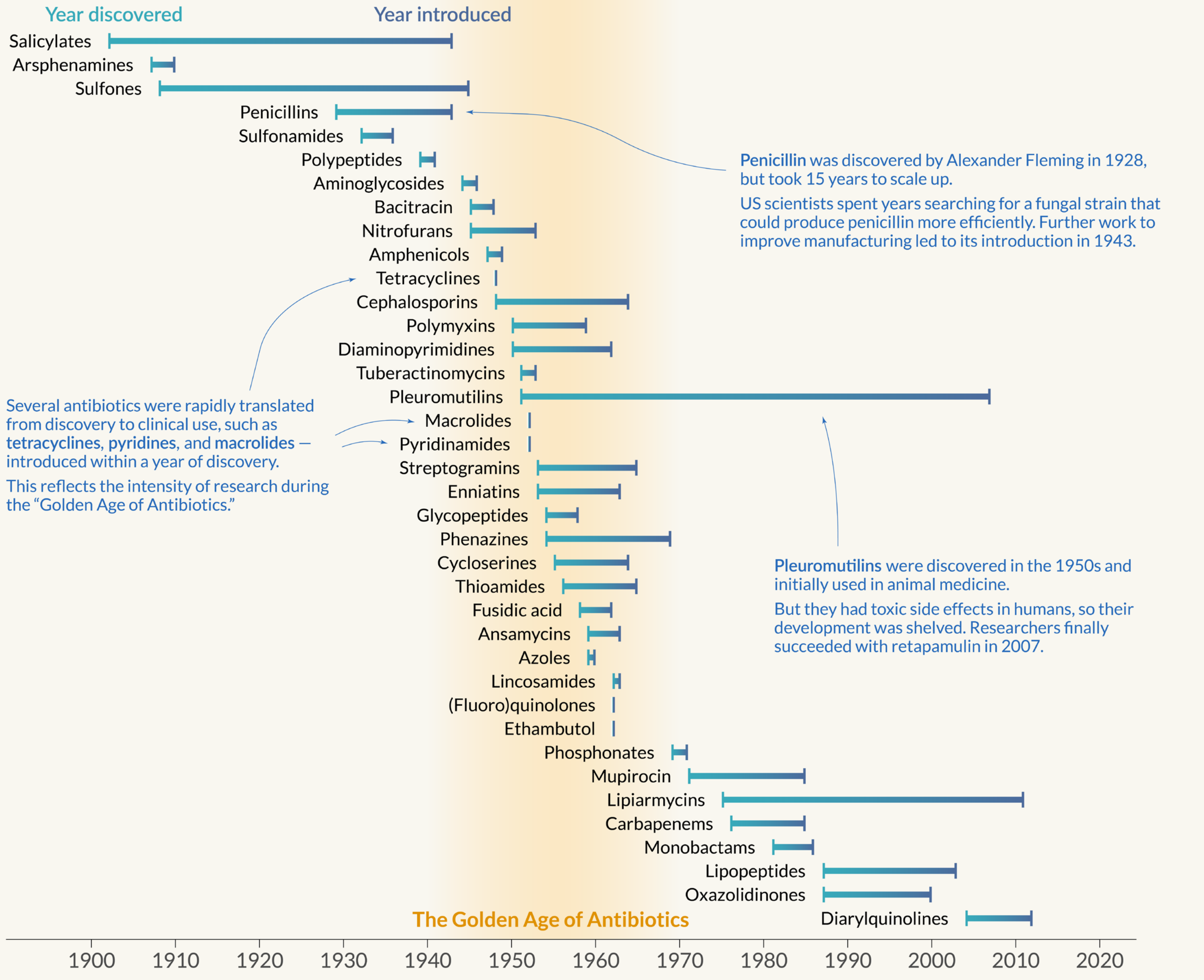
Cell wall synthesis



Antibiotics: time from discovery to introduction

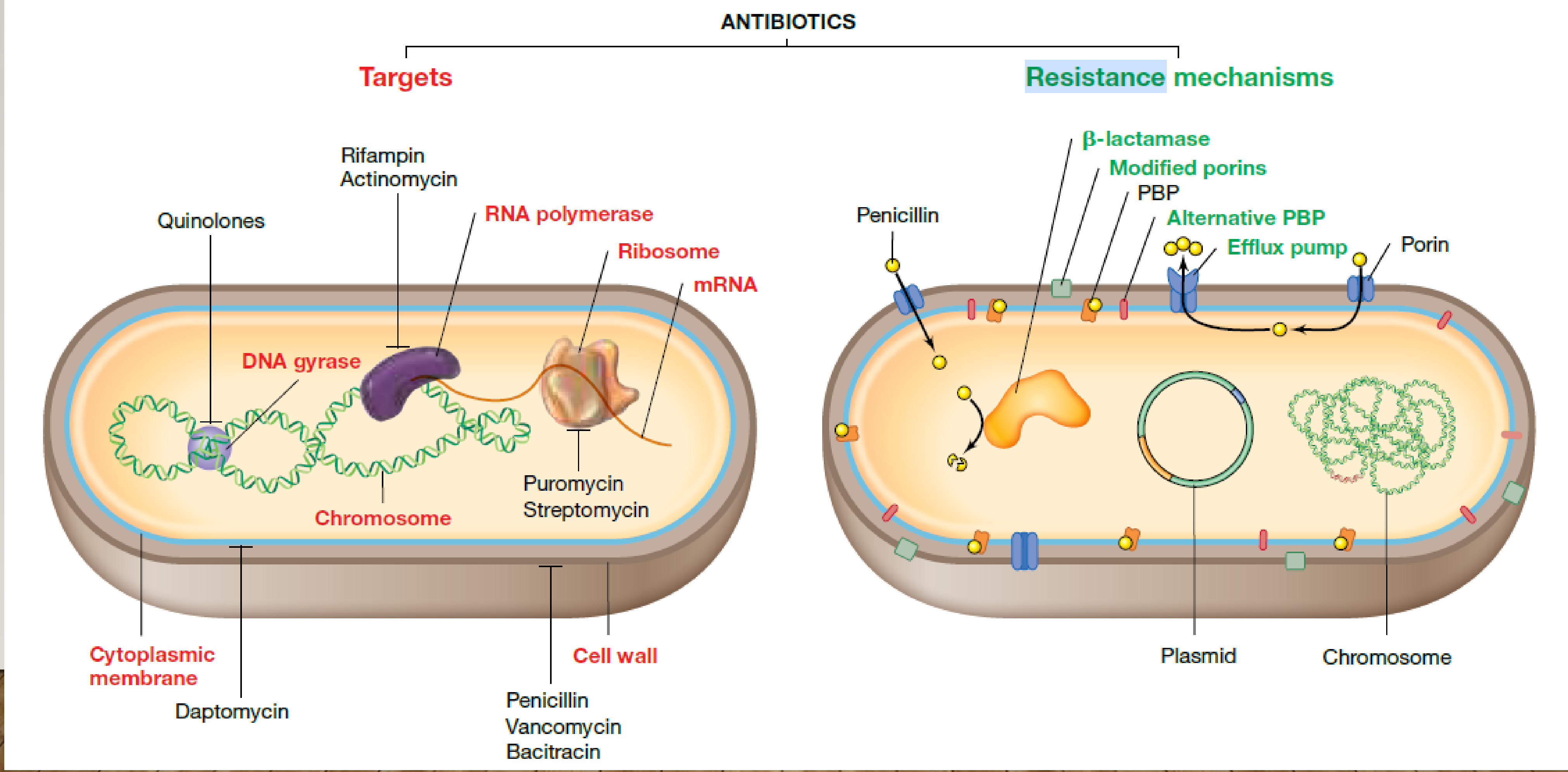
Our World in Data

The timespan between when each antibiotic drug class was discovered and when it was first in clinical use.

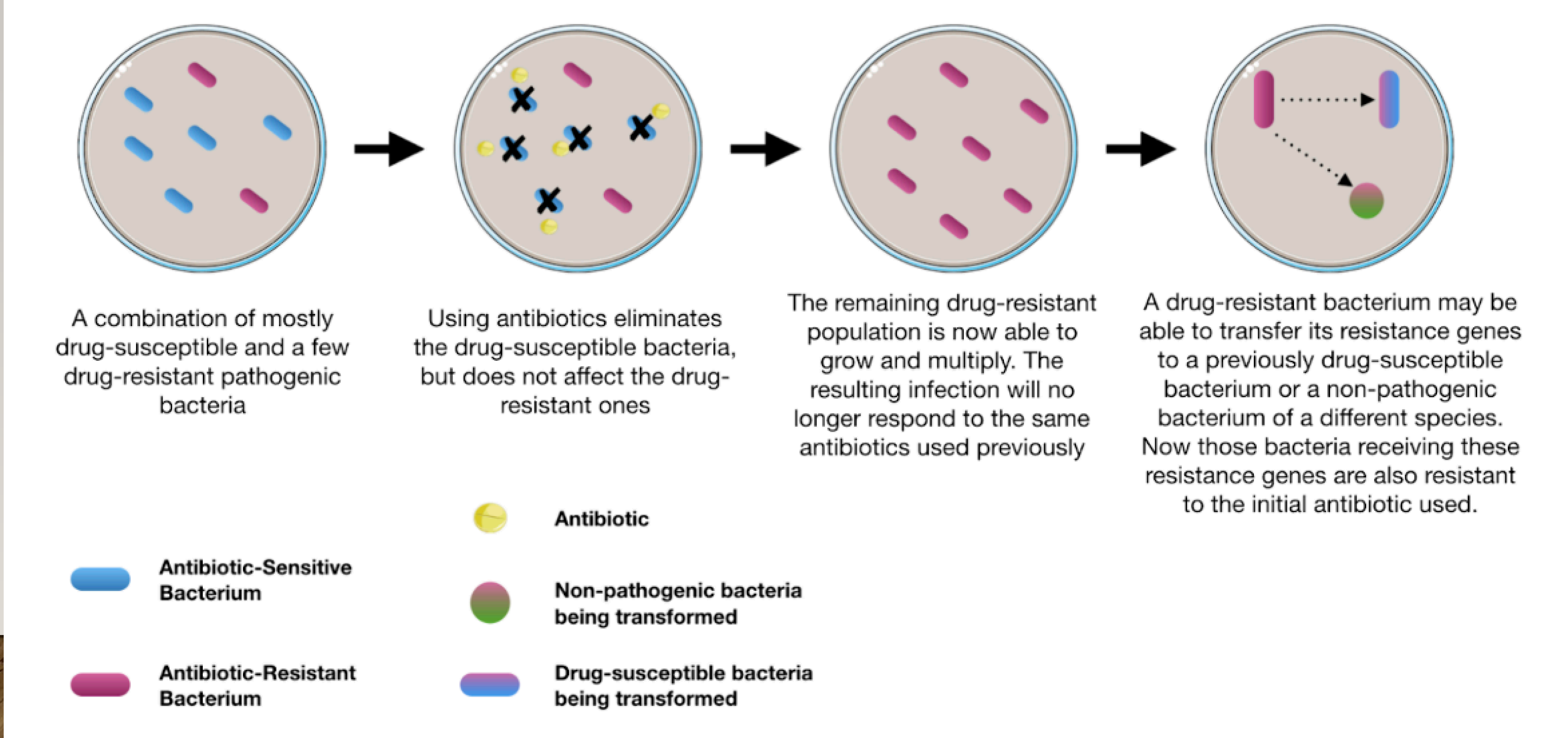


Source: Hutchings, Truman, Wilkinson (2019) Antibiotics: *Past, present and future.*
Only shown for antibiotic drug classes that have been introduced clinically. As of 2023, no new classes have become available; see AntibioticDB for more.
OurWorldinData.org — Research and data to make progress against the world's largest problems.

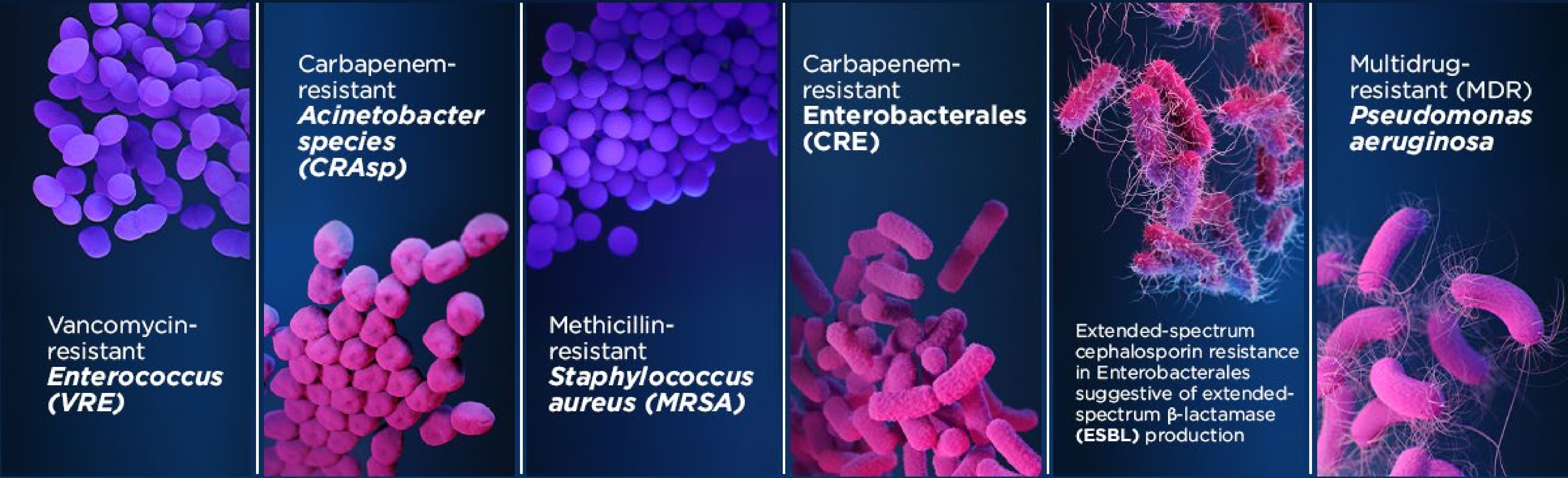
ANTIBIOTIC RESISTANCE



ANTIBIOTIC RESISTANCE



6 of the 18 most alarming **antibiotic resistance threats** cost the U.S. more than **$4.6 billion annually**



www.cdc.gov/DrugResistance

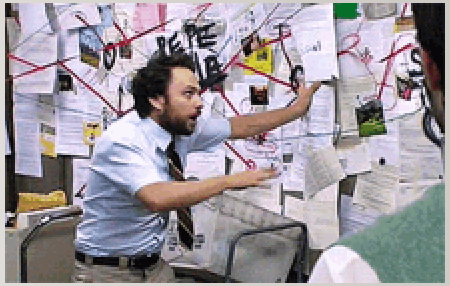
CDC

U.S. Department of Health and Human Services
Centers for Disease Control and Prevention

PROTOZOAN PATHOGENS

The protist kingdom

PROTISTS



- All eukaryotes except land plants, fungi and animals

can be a single cell to 200 meters long

Single cell

photosynthesis

Colonies

heterotrophs

Multicellular organisms

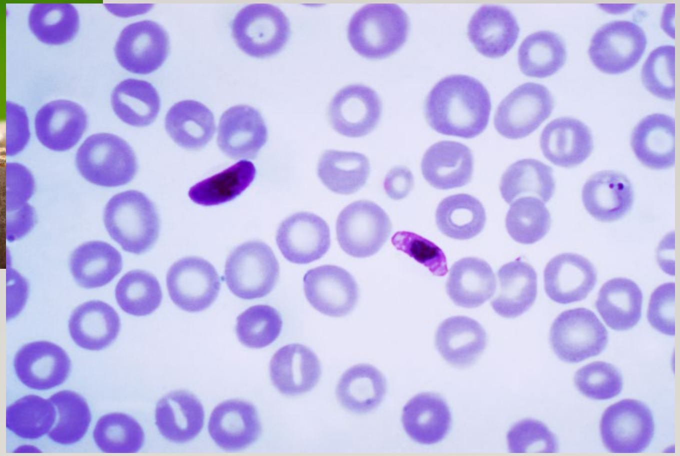
Some have hard shells, some resemble plants

Tend to live in environments where they are surrounded by water most of the time

PATHOGENIC PROTISTS

Species	Disease
Five species of Plasmodium, primarily P. falciparum and P. vivax	These organisms cause malaria. Over 3.4 billion people (roughly one-half of the world's population) live in areas at risk of **malaria** transmission
Naegleria fowleri	**Brain-eating amoeba** causes meningoencephalitis; infections are almost always fatal
Toxoplasma gondii	**Toxoplasmosis** may cause eye and brain damage in infants and in AIDS patients
Many species of dinoflagellates	Toxins released during harmful algal blooms accumulate in shellfish and poison people if eaten
Trypanosoma gambiense and T. rhodesiense	Trypanosomiasis ("**sleeping sickness**") is a potentially fatal disease transmitted through bites from tsetse flies. Occurs in Africa
Trypanosoma cruzi	**Chagas disease** affects 6–7 million people and causes 50,000 deaths annually, primarily in South and Central America
Entamoeba histolytica	Amoebic **dysentery** results in severe diarrhea and subsequent dehydration

Cases	Mortality
263 million	**8%**
estimated malaria cases in 83 malaria endemic countries in 2023	global decrease in the mortality rate since 2015



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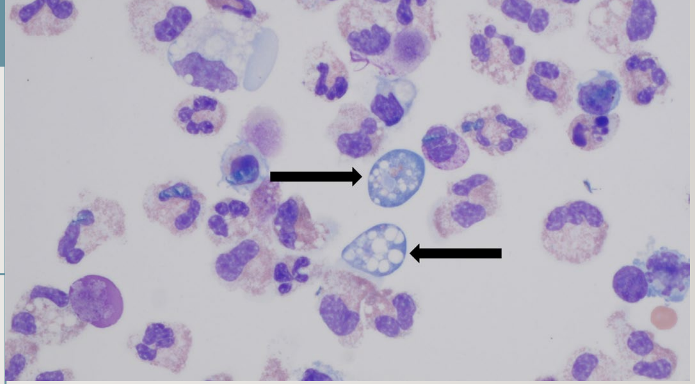


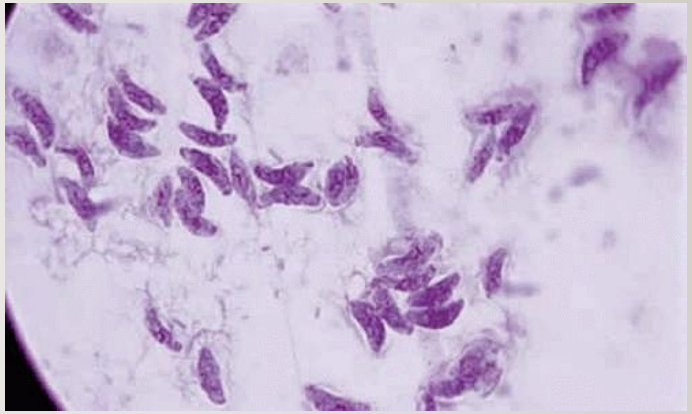
PATHOGENIC PROTISTS

Naegleria fowleri **infections are rare and devastating. From 2012 to 2021, 31 infections were reported in the U.S. All but three were fatal.**

You cannot be infected with *Naegleria fowleri* by drinking contaminated water, and the infection cannot spread from one person to another.

Infections are contracted through the nose.

PATHOGENIC PROTISTS



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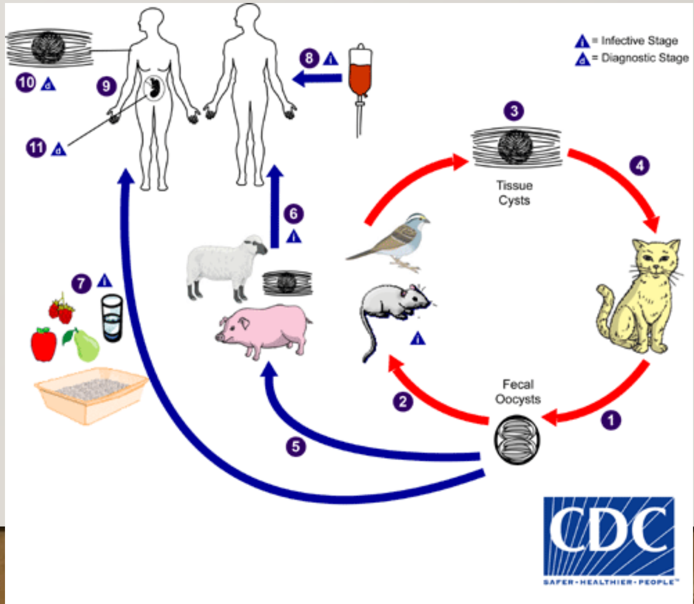

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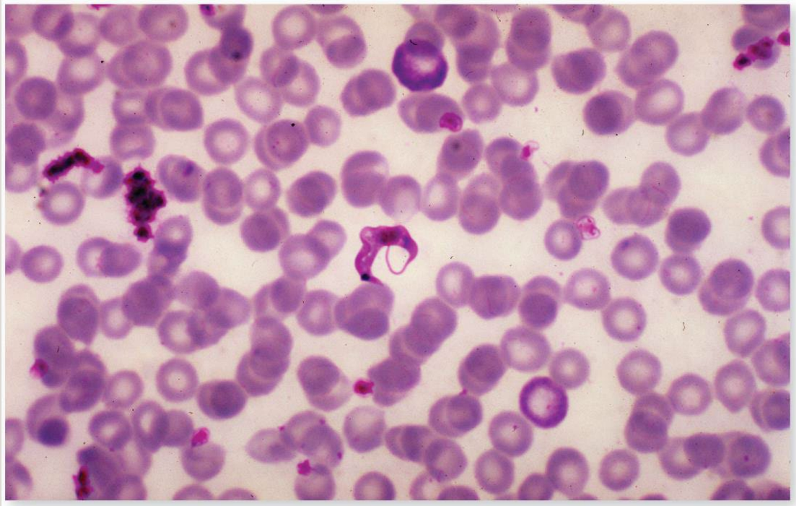

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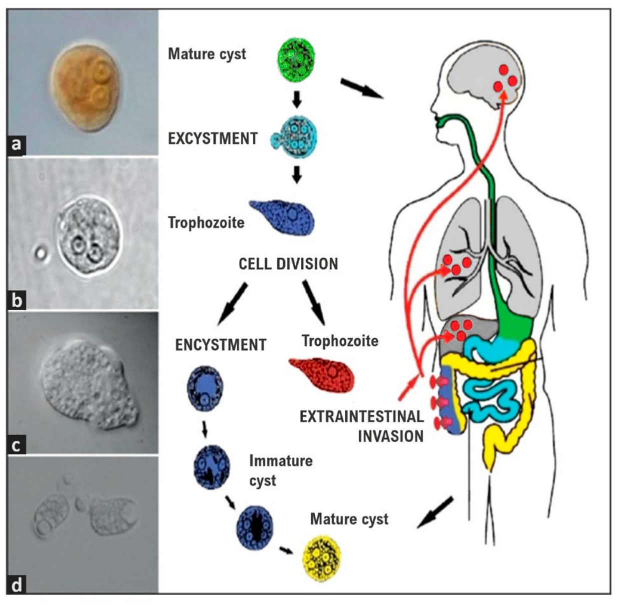


FUNGAL PATHOGENS

FUNGI

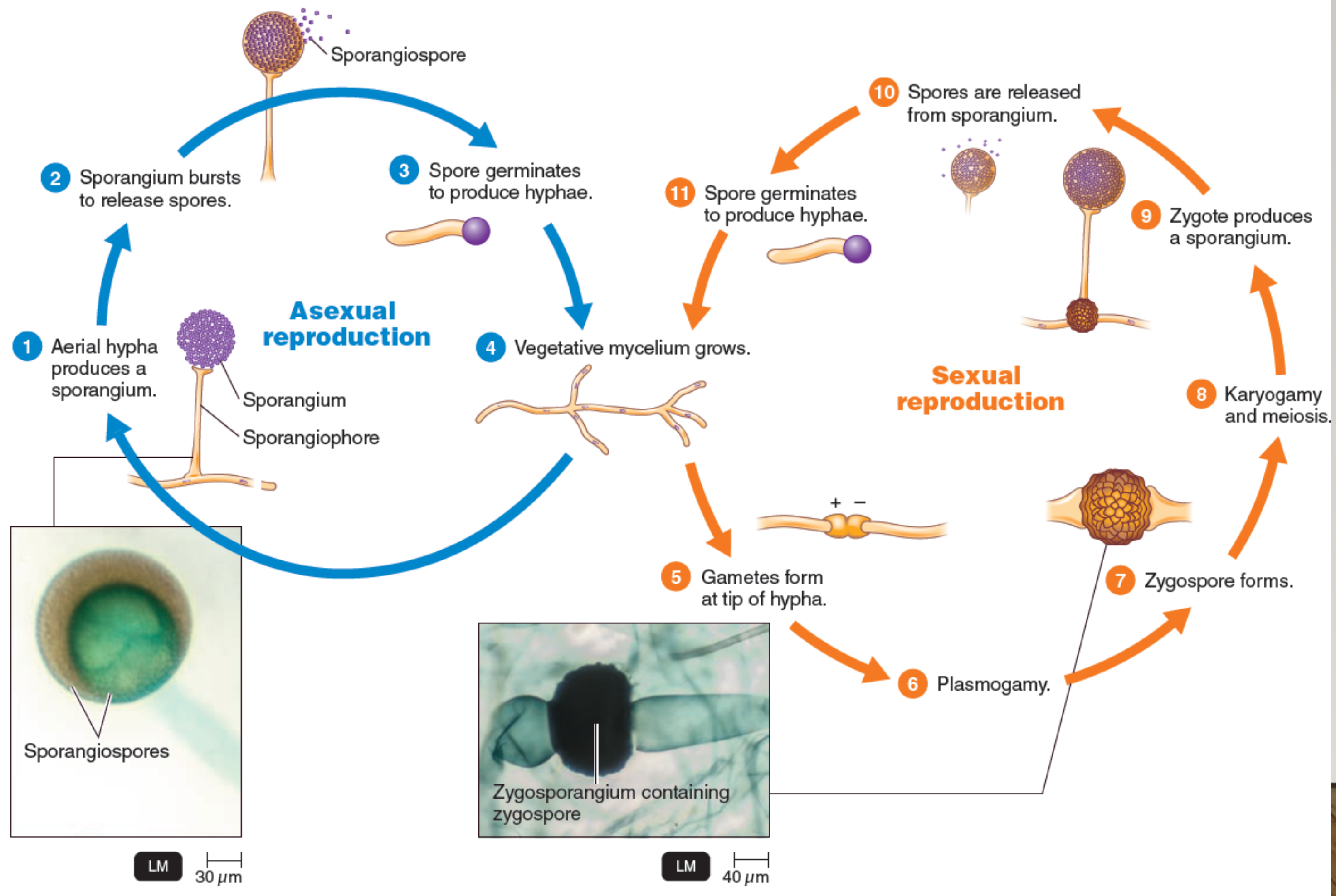
- Eukaryotes
- Heterotrophic: release digestive enzymes to absorbing small molecules
- Serve as major decomposers by absorbing nutrients from dead organisms
- Mutualists (interactions between individuals of different species that benefit both)

FUNGAL CHARACTERISTICS

- Yeast or mold

Single cell

Asexual budding



FUNGAL INFECTIONS

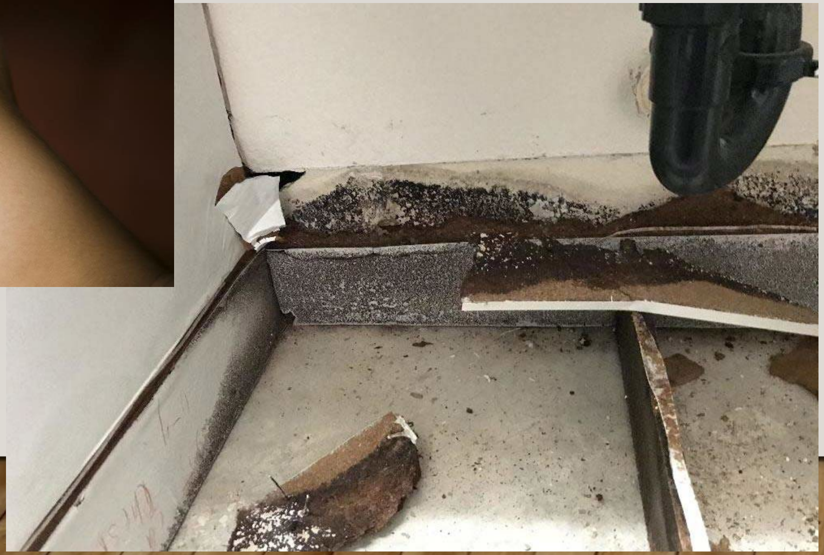


TABLE 33.1 Major pathogenic fungal diseases[a]

Class and disease	*Causal organism*	*Site*
Superficial mycoses		
Athlete's foot	*Epidermophyton, Trichophyton*	Between toes, skin
Jock itch	*Trichophyton, Epidermophyton*	Genital region
Ringworm	*Microsporum, Trichophyton*	Scalp, face
Subcutaneous mycoses		
Sporotrichosis	*Sporothrix schenckii*	Arms, hands
Chromoblastomycosis	*Phialophora verrucosa*, other fungi	Legs, feet, hands
Systemic mycoses		
Aspergillosis	*Aspergillus* spp.[b]	Lungs
Blastomycosis	*Blastomyces dermatitidis*	Lungs, skin
Candidiasis	*Candida albicans*[c]	Oral cavity, intestinal tract, vagina
Coccidioidomycosis	*Coccidioides immitis*[c]	Lungs
Paracoccidioidomycosis	*Paracoccidioides brasiliensis*	Skin
Cryptococcosis	*Cryptococcus neoformans*[c]	Lungs, meninges
Histoplasmosis	*Histoplasma capsulatum*[c]	Lungs
Pneumocystis pneumonia	*Pneumocystis jirovecii*[c]	Lungs

FOODBORNE FUNGAL ILLNESS

- Fungus good

- Fungus bad

(a) Parasitic fungi infect corn and other crops.

(b) Saprophytic fungi rot fruits and vegetables.

THE LAST OF FUNGUS

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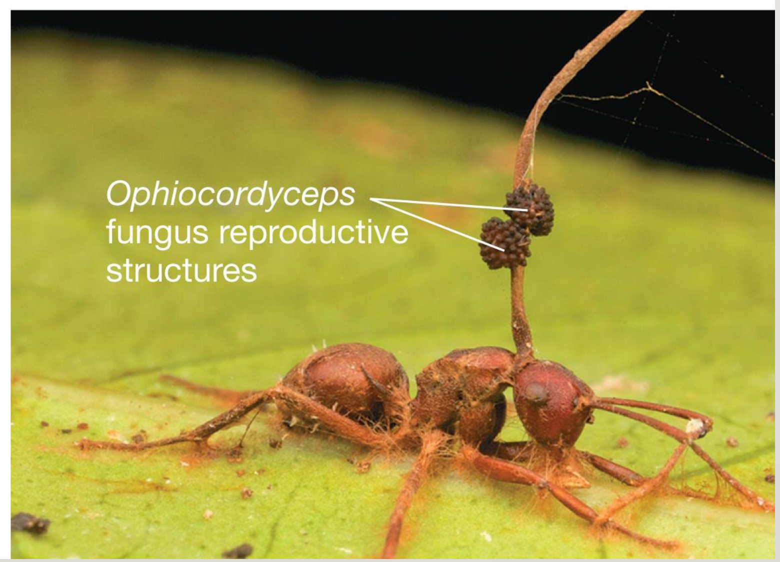
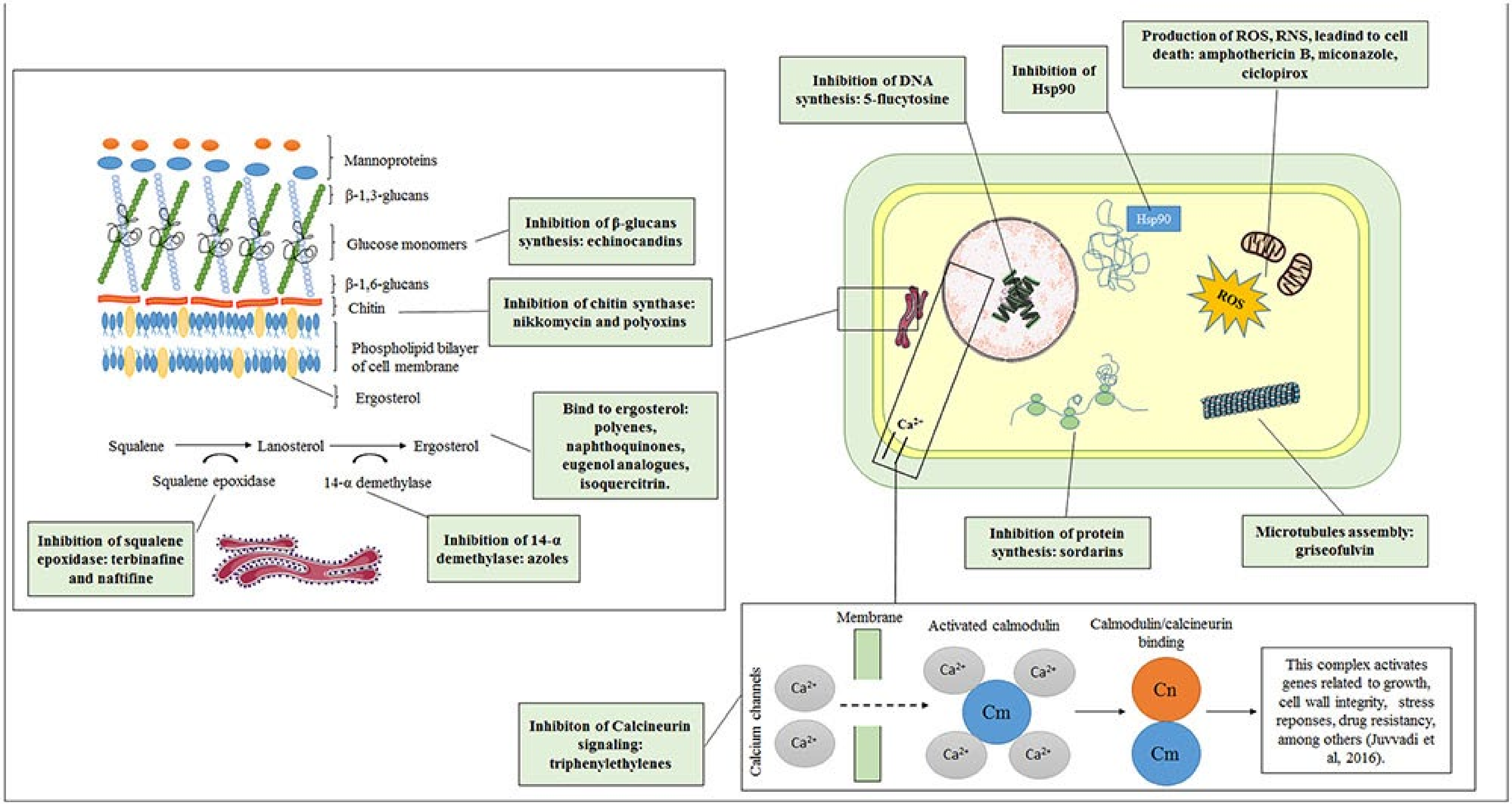


FUNGAL INFECTIONS - TREATMENT

	Mode of Action	Comments
ANTIFUNGAL DRUGS		
Agents Affecting Fungal Sterols (Plasma Membrane)		
Polyenes		
Amphotericin B	Injury to plasma membrane	Systemic fungal infections; fungicidal
Azoles		
Clotrimazole, miconazole	Inhibit synthesis of plasma membrane	Topical use
Ketoconazole	Inhibits synthesis of plasma membrane	Can be taken orally for systemic fungal infections
Allylamines		
Terbinafine, naftifine	Inhibit synthesis of plasma membrane	Treatment of diseases resistant to azoles
Agents Affecting Fungal Cell Walls		
Echinocandins		
Caspofungin (Cancidas)	Inhibits synthesis of cell wall	Intravenous use only
Agents Inhibiting Nucleic Acids		
Flucytosine	Inhibits RNA synthesis	Usually in combination with other antifungals
Other Antifungal Drugs		
Griseofulvin	Inhibition of mitotic microtubules	Fungal infections of the skin
Tolnaftate	Unknown	Athlete's foot

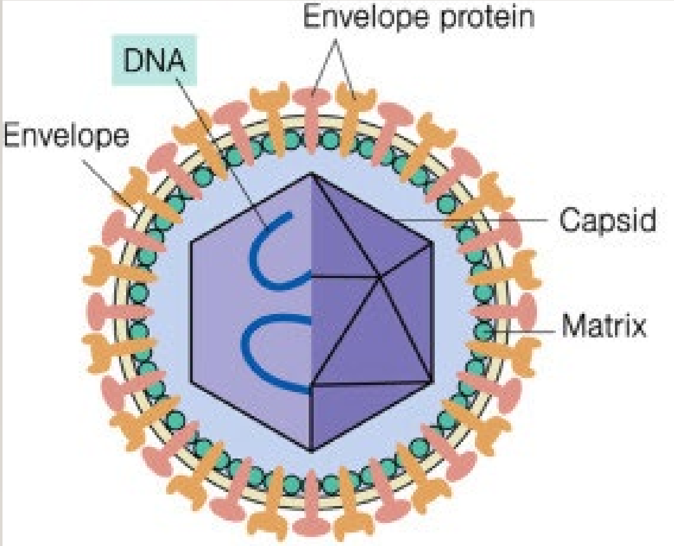
FUNGAL INFECTIONS - TREATMENT

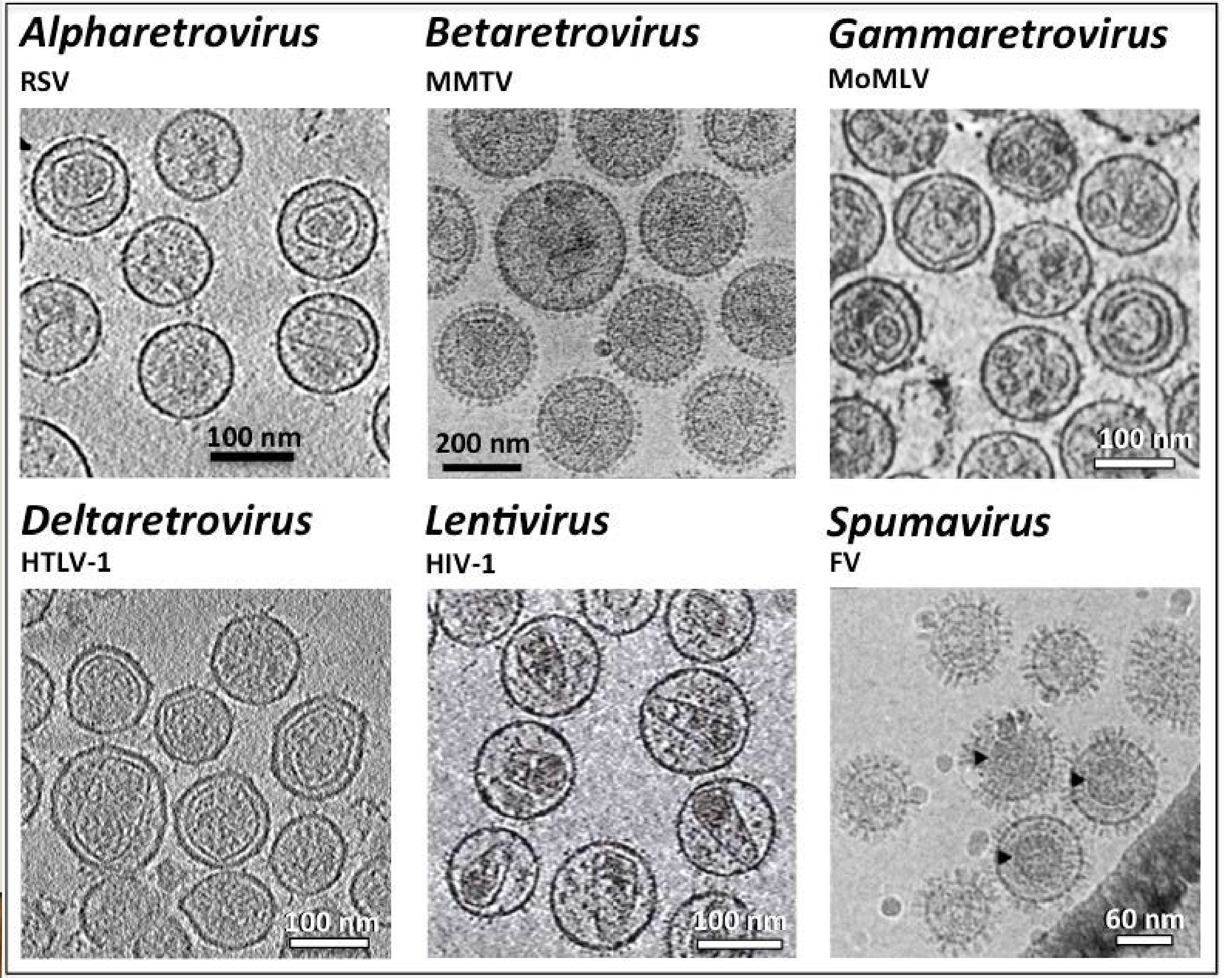


VIRUSES

WHAT ARE VIRUSES?

- Two main biomolecules
  - Proteins
  - Nucleic acids





https://www.scientificamerican.com/interactive/inside-the-coronavirus/

WHAT ARE VIRUSES?

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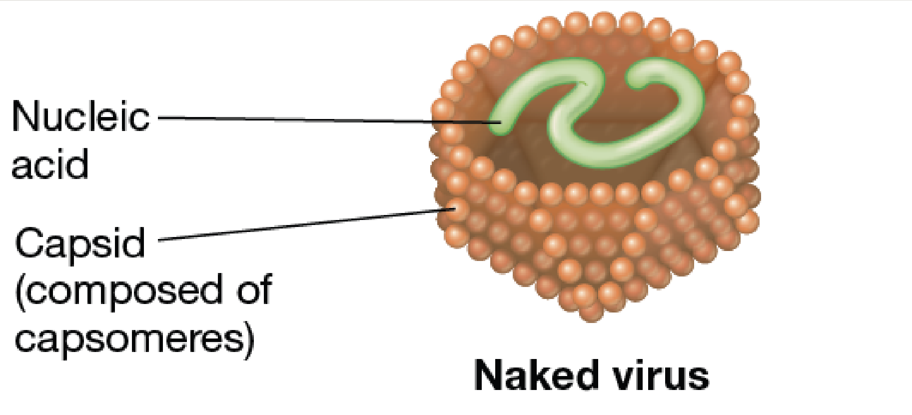
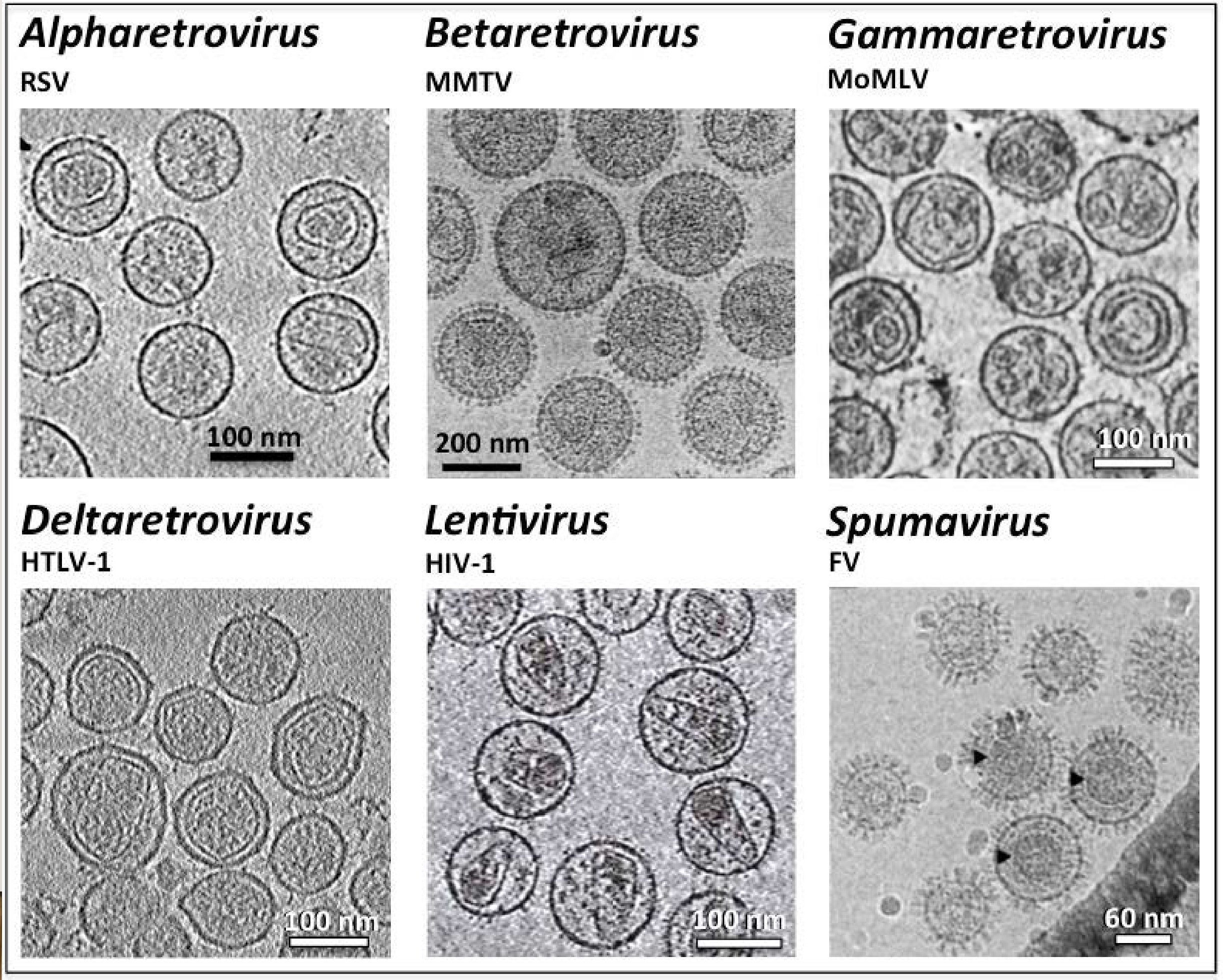
  - Nucleic acids

Essential components = Genome + Capsid





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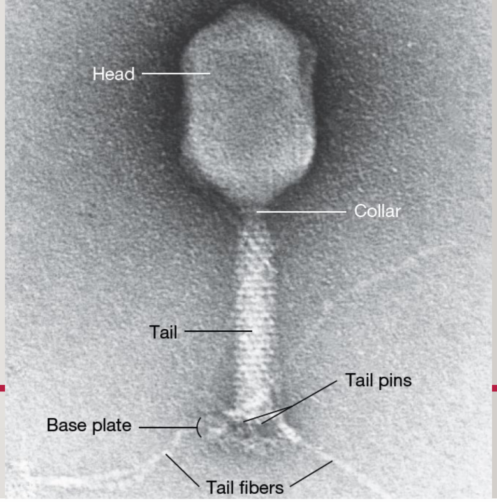
WHAT ARE VIRUSES?



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  - Proteins
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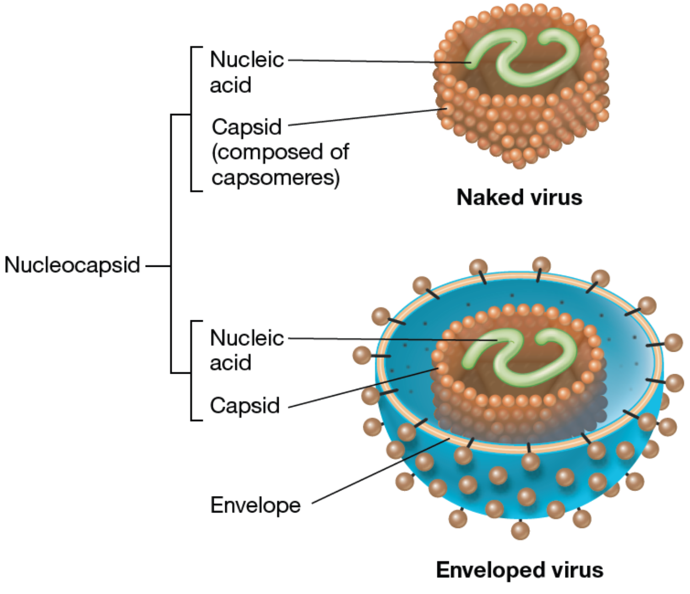

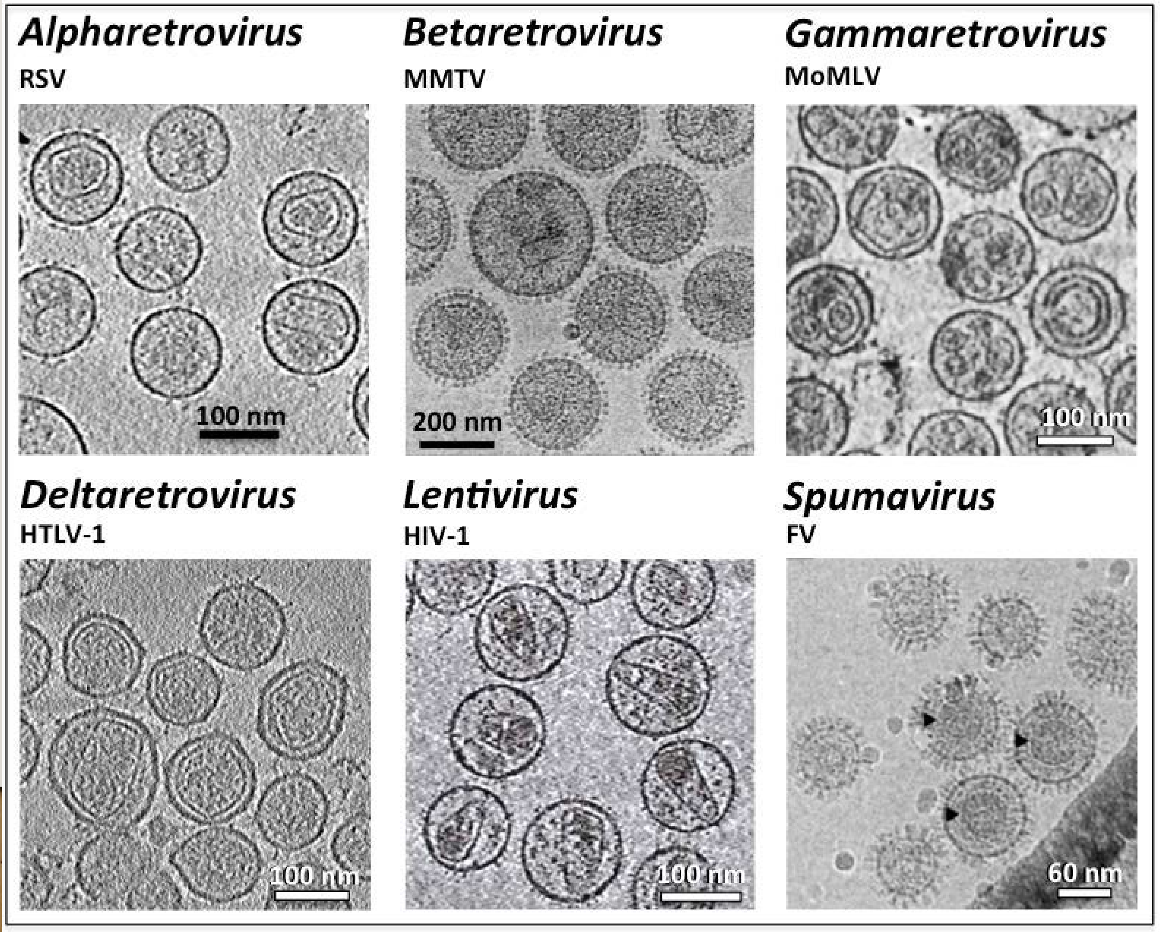

Essential components = Genome + Capsid

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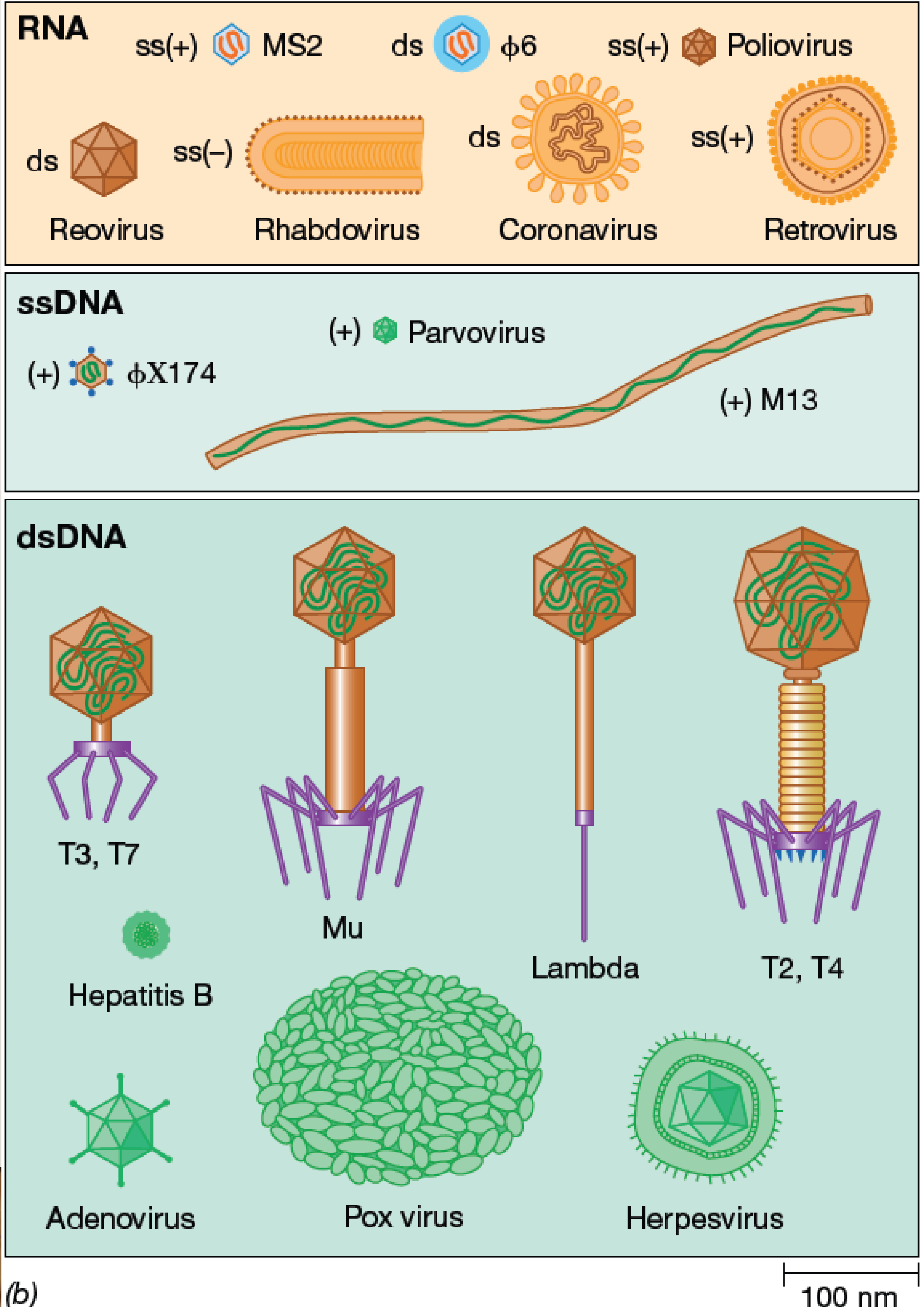
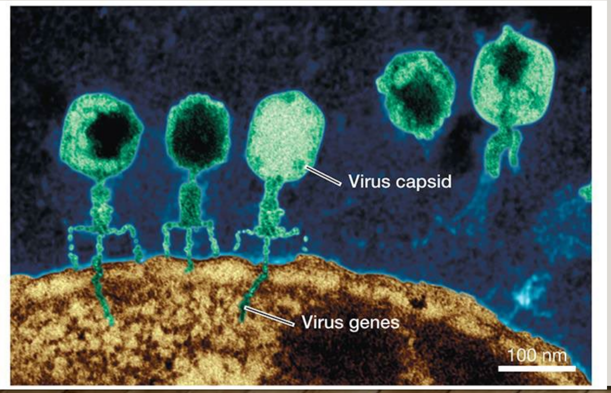
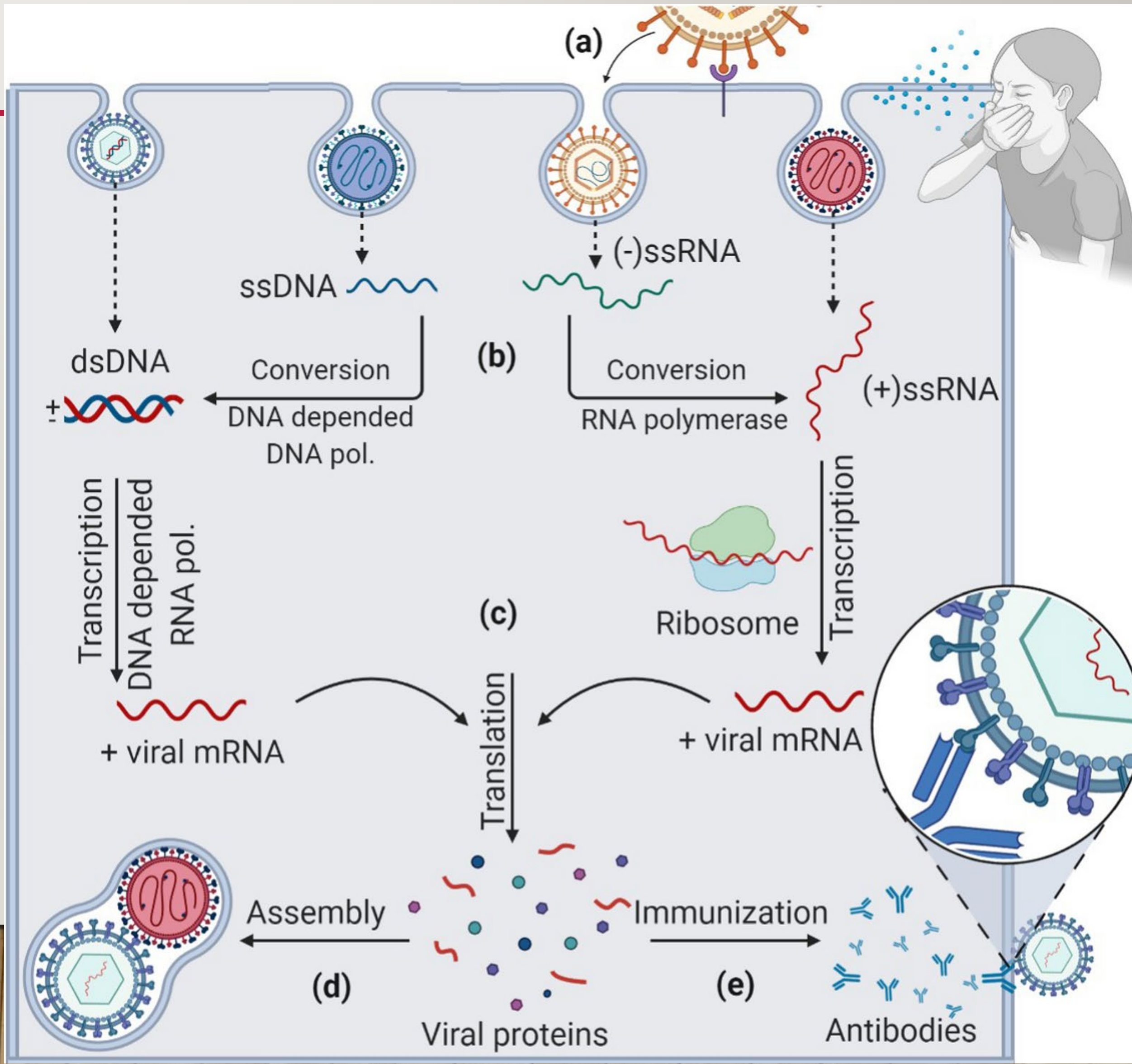
VIRAL GENETIC MATERIAL



VIRUSES NEED CELL MACHINERY

1. Injecting genetic material
2. Using host cell for transcription and translation
3. Assembling proteins and genetic material into new copies of the virus

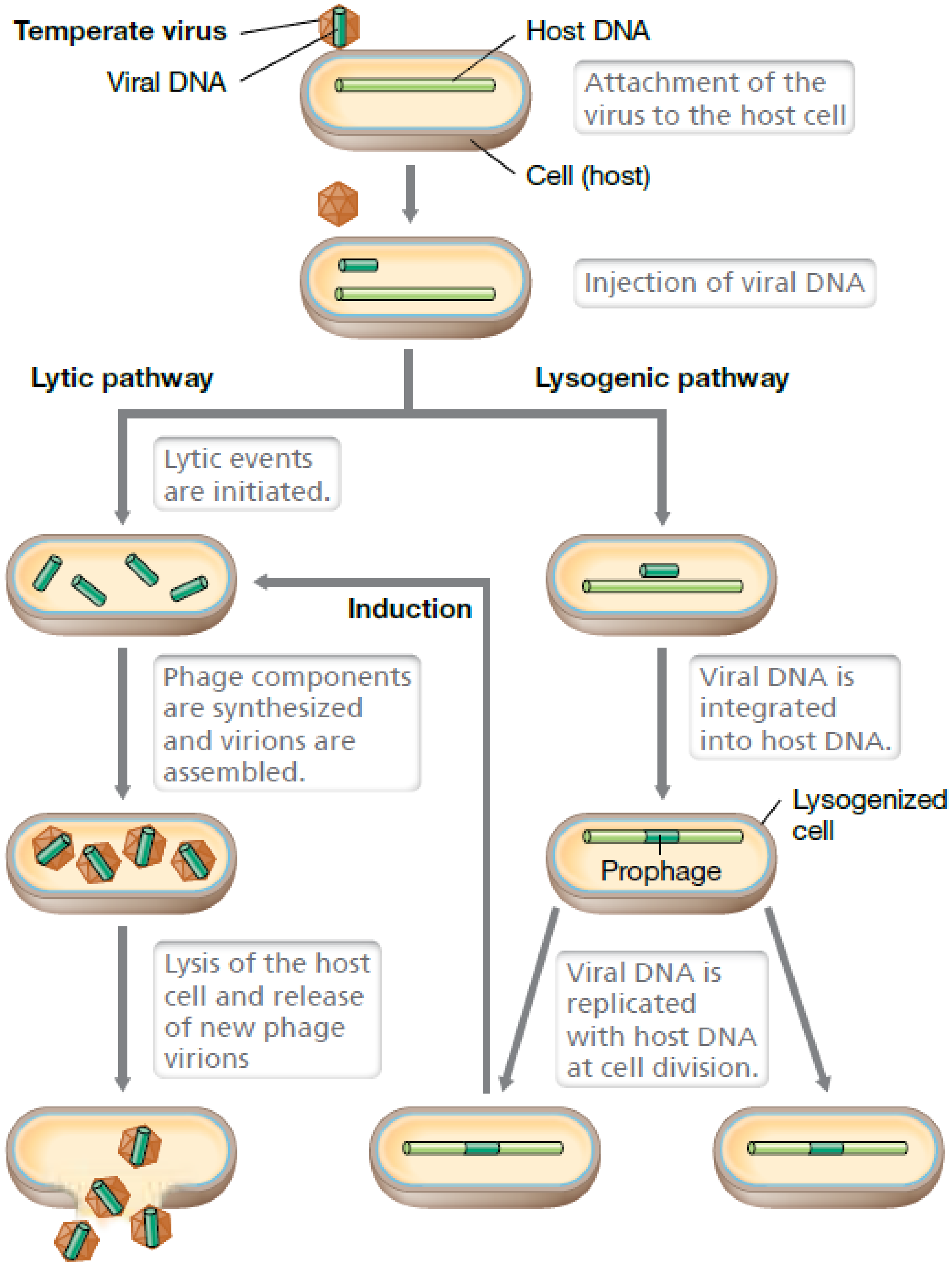




VIRUS "LIFE" CYCLE

- In the **Lytic** cycle:
  - a virus begins replicating almost immediately upon entry into the cell.
  - Its genome is independent of the host genome
  - Most viruses use the lytic cycle as their default life cycle

- In the **Lysogenic** cycle:
  - The genome incorporates with the host genome but remains inactive
  - Viral reproduction will not occur until an environmental signal induces the viral DNA to be read and processed
  - HIV, Herpes Simplex, and some Hepatitis viruses



COMING UP…

- First Intentional Scientific Attempts to Control Infectious Disease

STUDY GUIDE

a) Describe what makes something pathogenic.

b) List the 4 classes of pathogens we discussed in class.

c) Be able to identify each class of pathogen based on structural descriptions

d) Describe how viruses survive and reproduce.

e) Describe generally how antibiotics and antifungals work and how resistance can develop